

Modelling with the Reaction-Diffusion Equation

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Preface

One purpose of mathematical modelling is to obtain a deeper understanding of a system or a phenomenon. The new insight will, hopefully, enable more precise system predictions and control. Mathematical modelling can also motivate one to acquire new mathematical skills, and this gives rise to explorations of fundamental mathematics.

This text seeks to give an introduction to mathematical modelling of reaction-diffusion systems. The focus is on phenomenology and not on providing detailed quantitative models of specific problems. In the exploration we will need knowledge of different topics, for example,

- boundary value problems,
- series solutions,
- the regular perturbation method,
- Fourier series and the Fourier transform,
- the diffusion equation,
- some particular techniques to analyse the reaction-diffusion equation,
- and simple numerical methods.

The text will introduce these topics. The numerical algorithms are found in the appendices and code can be downloaded from the course moodle webpage.

There are, of course, many different ways we can approach our exploration, but there are some obvious choices based on *the principle of progression*. For example, we can use some important results from pure diffusion problems to solve the linear reaction-diffusion equation, hence, it will be meaningful to treat the former first. On this basis, the text is based on three main themes, namely,

the steady state $\rightarrow$ the diffusion equation $\rightarrow$ the react. diff. equation.

The arrows indicate dependencies.

Our treatment is based on examples, questions, exercises, and, to some extent, proofs of theorems. It is not the purpose here to give a formal theoretical foundation of the different topics; this calls for a much more rigorous approach and requires advanced knowledge of, for example, functional analysis. The theorems will some times be stated under "mild" conditions and the proofs will be of the "calculus"-type. In this way it follows the standard format for many popular texts on mathematical biology. I will refer the interested reader to the more advanced literature whenever I can.

The prerequisite needed to read the note is (i) calculus and basic linear algebra, (ii) ordinary differential equations and initial value problems, and (iii) dynamical systems analysis (e.g. phase portrait analysis). Basic knowledge of real analysis can be helpful.

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Chapter 1

Introduction

We dive right in: Consider a thin tube with cross section area A and length L . The longitudinal axis is parallel to the x -axis as shown in Fig. 1.1. Inside the tube there is a bio-chemical substance, S , and we denote the concentration (or density) of S by c . The concentration is a function of position, x , and time, t , and the fundamental research question throughout this text is

how does the function $c = c(x, t)$ evolve in space and time considering diffusion and reaction processes?

Why this question is interesting in the first place will hopefully be clear as we explore the topic in this and the following chapters.

Now, the first step is to derive the underlying dynamical equation for c ; we do this using a rather intuitive approach. Consider a slice volume of the tube which is confined between x_1 and x_2 . Define the width of the slice $\Delta x = x_2 - x_1$. The slice volume is then $\Delta V = A\Delta x$. By the *flow rate* Q at x_1 we mean the amount of S leaving or entering the slice volume at $x = x_1$ per unit time. Likewise, the flow rate at x_2 is the amount of S leaving or entering the slice at $x = x_2$ per unit time.

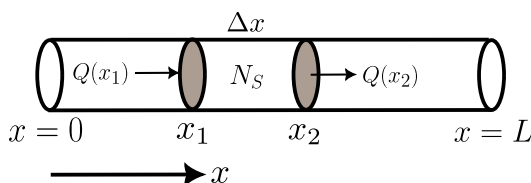


Figure 1.1: Schematic illustration of the effective one-dimensional tube system. The shaded areas are the cross-sectional areas of the tube. See the text for explanations of the symbols.

There is an important convention here: The flow rate is positive when the substance flows to the right (in the “positive” x -direction) and it is negative if it flows to the left (“negative” x -direction).

S can also be produced or consumed inside the slice, for example, through biochemical processes. The production (or consumption) per unit time in the volume we denote σ_S . If N_S is the total amount of S in the slice the rate of change of S is written as

$$\begin{aligned}\frac{dN_S}{dt} &= \text{production rate} + \text{flow rate at } x_1 - \text{flow rate at } x_2 \\ &= \sigma_S(t) + Q(x_1, t) - Q(x_2, t).\end{aligned}\quad (1.1)$$

This equation is the *balance equation* for S in the slice. The amount of S is related to the average concentration in the slice c_S by

$$N_S = \Delta V c_S \quad \text{implying} \quad \frac{dN_S}{dt} = \Delta V \frac{dc_S}{dt}.\quad (1.2)$$

Substitution into Equation (1.1) and dividing by the volume we get

$$\frac{dc_S}{dt} = \frac{\sigma_S(t)}{\Delta V} - \frac{1}{A} \frac{Q(x_2, t) - Q(x_1, t)}{\Delta x}.\quad (1.3)$$

Now, $x_2 = x_1 + \Delta x$ and we can find the rate of change at the point x_1 by letting $\Delta x \rightarrow 0$. In this limit the first term on the right-hand side of Equation (1.3) we denote the *production function* at x_1 and time t . The second term is just a partial derivative

$$\left. \frac{\partial J(x, t)}{\partial x} \right|_{x=x_1} = \frac{1}{A} \lim_{\Delta x \rightarrow 0} \frac{Q(x_1 + \Delta x, t) - Q(x_1, t)}{\Delta x}.\quad (1.4)$$

J is the *flux* and is the flow rate per unit area, $J = Q/A$. We can do this for any interior point in the tube, that is, for any $x \in]0; L[$ we have that

$$\frac{\partial}{\partial t} c(x, t) = f(x, t) - \frac{\partial}{\partial x} J(x, t),\quad (1.5)$$

Equation (1.5) is only meaningful if both c and J have partial derivatives in their domain.

Equation (1.5) formalizes that the local rate of change of the concentration of S at any point along the tube changes due to (i) local production or consumption and (ii) the spatial variation of the flux. As it stands it does not form a closed problem, since we do not know f or J ; these we must model.

If the production is due to bio-chemical reactions we can write the production function in terms of the *reaction function*, $r(c) = r(c(x, t)) = f(x, t)$.

In general, the flux J can be decomposed into a term due to *advection* and a term due to *diffusion*. Advection is the process that transports a substance around the system when there is a bulk flow; we will not consider this process.

Diffusion is the process that tends to remove any gradient in the system: If you add a drop of coloured dye to a glass of water the dye molecules will eventually distribute uniformly in the system. Therefore, we intuitively expect that the flux due to diffusion is non-zero whenever there exists a gradient in the concentration. This is expressed in *Fick's first law*

$$J = -D \frac{\partial c}{\partial x}.\quad (1.6)$$

Here $D > 0$ is the *diffusion coefficient*, which in general is dependent on temperature, position, and so forth. If $\partial c / \partial x < 0$ then $J > 0$ and there is a flux from left to right due to diffusion, and vice versa, if $\partial c / \partial x > 0$ then $J < 0$ and the flux is from right to left.

If we substitute $f(x, t) = r(c)$ and Fick's law into Equation (1.5) we get the *reaction-diffusion equation*

$$\frac{\partial c}{\partial t} = r(c) + \frac{\partial}{\partial x} \left[D \frac{\partial c}{\partial x} \right]. \quad (1.7)$$

We abbreviate the reaction-diffusion equation by *RD-equation*. If the diffusion coefficient is constant the RD-equation becomes

$$\frac{\partial c}{\partial t} = r(c) + D \frac{\partial^2 c}{\partial x^2}. \quad (1.8)$$

Importantly, Fick's law is a model and it is therefore, perhaps, unfortunate that the literature refers to this as a "law"¹. It proposes that the flux is linearly dependent on the substance concentration gradient, that is, it is the first-order term in the Taylor expansion about zero concentration gradient. This type of model is more formally referred to as a *linear constitutive relation*.

The RD-equation can be used to model many systems other than bio-chemical systems, and c can represent fluid velocity, temperature, the concentration of large animals, and much more. We will also encounter examples of this.

Example 1 An important example we will investigate throughout the text is the case of a simple linear growth rate of, say, some biological organism like plankton. In addition to growth, we also include the effect of diffusion in one direction (we have the plankton in a small radius tube). If c denotes the organism concentration and if the diffusion coefficient is constant we have from Equation (1.8)

$$\frac{\partial c}{\partial t} = kc + D \frac{\partial^2 c}{\partial x^2}, \quad (1.9)$$

where k is the growth rate coefficient. If the population grows $k > 0$; if it declines $k < 0$. This RD-equation is a linear differential equation and we should be optimistic about finding a solution.

The RD-equation must be solved by specifying both an *initial condition* (IC) and *boundary conditions* (BCs). If the organism at time zero is bundled up around some interior point $x_0 \in]0; L[$, we may specify the IC as a Gaussian-shaped curve

$$c(x, 0) = c_0 e^{-(x-x_0)^2/\sigma^2}, \quad (1.10)$$

where c_0 is the maximum concentration and σ determines the width of the Gaussian curve. If we somehow can control the concentrations at the tube ends $x = 0$ and $x = L$ such that it is zero at these points, we write the BCs as

$$c(0, t) = 0 \quad \text{and} \quad c(L, t) = 0 \quad \text{for} \quad t \geq 0. \quad (1.11)$$

¹The same can be said about Ohm's law, but *not* e.g. Newton's second law

These two BCs define a specific function value at the system endpoints, and are called *Dirichlet boundaries* or boundary conditions of the first type. In general, the boundary value can take any (real) value and not just zero. The boundary at $x = 0$ we will refer to as the *left boundary* and the boundary at $x = L$ the *right boundary*.

Another commonly used BC is the *Neumann boundary*, or boundary condition of the second type. This specifies the values for the function derivative at the walls, for example,

$$\left. \frac{\partial c}{\partial x} \right|_{x=0} = 0 \quad \text{and} \quad \left. \frac{\partial c}{\partial x} \right|_{x=L} = 0 \quad \text{for } t \geq 0. \quad (1.12)$$

We can also encounter systems with *mixed* BCs, such that at, say, the left boundary we have a Neumann BC and at the right boundary we have a Dirichlet BC

$$\left. \frac{\partial c}{\partial x} \right|_{x=0} = 0 \quad \text{and} \quad c(L, t) = C_L \quad \text{for } t \geq 0. \quad (1.13)$$

We shall see that the BCs play a crucial role in determining both the form of the solution and, in some cases, whether a solution exists. $\diamond$

Question What is the bio-physical interpretation of the Neumann BC given in Equation (1.12)?

Example 2 The RD-equation is the same equation used to describe fluid flows in some special – yet important – cases. Let a fluid be confined between two parallel walls located at $x = 0$ and $x = L$. For sufficiently small Reynolds number (for example, for sufficiently small velocities), the inertial term may be neglected, and we obtain

$$\rho \frac{\partial v}{\partial t} = \rho g + \eta_0 \frac{\partial^2 v}{\partial x^2}, \quad (1.14)$$

where v is the flow in the y -direction, the production function, ρg , is the local applied force generating the flow, and η_0 is the viscosity. The viscosity can be interpreted as the diffusion coefficient for fluid momentum.

If the applied force is constant the fluid flow will reach a *steady-state* after some time

$$\frac{\partial v}{\partial t} = 0. \quad (1.15)$$

We write this as $v(x, t) = v(x)$. In the steady-state we then have

$$\frac{d^2 v}{dx^2} = -\frac{\rho g}{\eta_0}. \quad (1.16)$$

This problem does not depend on the initial conditions and defines a *boundary value problem* which is a class of very important problems.

Consider the case where we do not apply a force, $g = 0$, and we move one wall with a speed V_0 while keeping the other fixed. If we apply Dirichlet boundaries we obtain the boundary value problem

$$\frac{d^2v}{dx^2} = 0 \quad (1.17)$$

with

$$v(0) = 0 \quad \text{and} \quad v(L) = V_0. \quad (1.18)$$

This is known as a Couette flow.

In a Poiseuille flow we apply an acceleration $g \neq 0$, but keep the walls fixed. Then we need Equation (1.16) with BCs

$$v(0) = 0 \quad \text{and} \quad v(L) = 0. \quad (1.19)$$

◇

Example 3 In bio-chemical systems there will be more than just one animal species, organism, or chemical compound that multiply, die, or react. In general, there will be very many "substances", $S_1, S_2, \dots$, and the corresponding concentrations, $c_1, c_2, \dots$, will be coupled. For example, if we wish to model a predator population we usually also need to include the food resources (prey) unless it is abundant and can be considered constant. The prey dynamics itself depends on the predator population as prey is eaten/consumed by predators, hence, the two are coupled.

If we have two coupled concentrations and assume that diffusion of c_1 only depends on the gradient of c_1 and that the diffusion of c_2 only depends on c_2 we can write a system of RD-equations

$$\frac{\partial c_1}{\partial t} = r_1(c_1, c_2) + D_1 \frac{\partial^2 c_1}{\partial x^2} \quad (1.20)$$

$$\frac{\partial c_2}{\partial t} = r_2(c_1, c_2) + D_2 \frac{\partial^2 c_2}{\partial x^2} \quad (1.21)$$

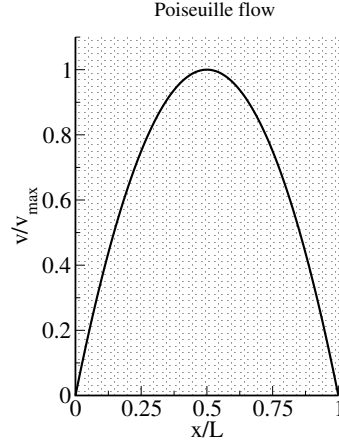
In short-hand vector notation we write this as

$$\frac{\partial \mathbf{c}}{\partial t} = \mathbf{R}(\mathbf{c}) + \mathbf{D} \frac{\partial^2 \mathbf{c}}{\partial x^2}, \quad (1.22)$$

where $\mathbf{c} = (c_1, c_2)$, $\mathbf{R}$ is a vector-valued reaction function $\mathbf{R} = (r_1, r_2)$ and $\mathbf{D}$ is the so-called diffusion matrix

$$\mathbf{D} = \begin{bmatrix} D_1 & 0 \\ 0 & D_2 \end{bmatrix}. \quad (1.23)$$

To proceed we again need to model the reaction terms r_1 and r_2 , as well as define the IC and BCs for both concentrations c_1 and c_2 .



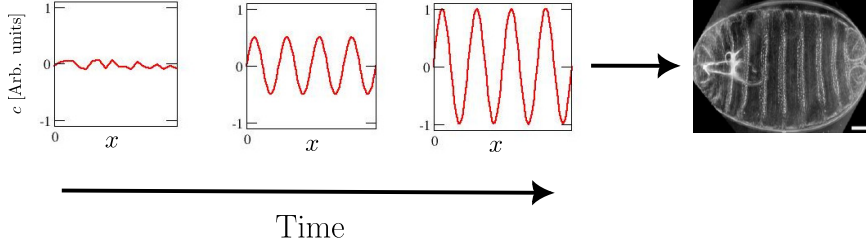
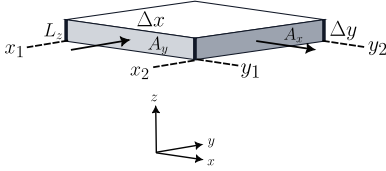


Figure 1.2: Turing structure. From an initially almost homogeneous state with small random perturbations (a), a spatial structure with a characteristic wavelength emerges and grows in amplitude (b), eventually approaching a stationary structure (c).

Alan Turing was the first to apply this type of model to account for *biological morphology*, that is, what appears to be spontaneously emerging structures in biological systems. If the system is initialized in a random (macroscopically uniform) fashion, see Fig. 1.2, then concentration gradients emerge spontaneously giving rise to a static (time independent) structure with respect to the concentrations. $\diamond$

Example 4 In the laboratory bacteria can be grown by seeding a small amount of bacteria onto a surface with nutrients and agar; often this is done in a Petri dish. As the bacteria multiply the colony grows radially from the seeding point.

The growth happens, effectively, in two spatial dimensions which we can highlight by writing



$$c = c(x, y, t). \quad (1.24)$$

We must extend the balance equation to describe this two dimensional situation, and we will do this by applying the same intuitive approach we did for one dimension. Let N_s be the total amount of the substance S in some volume given by $\Delta V = \Delta x \Delta y L_z$. We ignore flow of S in the z -direction, however, we do have flows in both the x - and y -directions. $A_x = \Delta y L_z$ is the area of the surface normal to the x -axis and $A_y = \Delta x L_z$ the area of the surface normal to the y -axis. The *total rate* through surface area A_x at $x = x_1$ is written as $Q_x(x_1, t)$ and at $x = x_2$ we write $Q_x(x_2, t)$. Equivalently, total flow rate through the surface areas A_y at $y = y_1$ and $y = y_2$ are $Q_y(y_1, t)$ and $Q_y(y_2, t)$, respectively. The balance equation states

$$\frac{dN_s}{dt} = \sigma_s(t) + Q_x(x_1, t) - Q_x(x_2, t) + Q_y(y_1, t) - Q_y(y_2, t) \quad (1.25)$$

We use the same convention for the flow rates as in one dimensions: The flow is positive if S flows in positive direction and negative if S flows in the negative direction.

Now, the concentration is simply $N_s(t) = \Delta V c(t)$, that is, the balance equation becomes

$$\frac{dc}{dt} = \frac{\sigma_S(t)}{\Delta V} - \frac{1}{A_x} \frac{Q_x(x_2, t) - Q_x(x_1, t)}{\Delta x} - \frac{1}{A_y} \frac{Q_y(y_2, t) - Q_y(y_1, t)}{\Delta y} \quad (1.26)$$

Again we write $x_2 = x_1 + \Delta x$ and $y_2 = y_1 + \Delta y$, and the fluxes J_x and J_y are now defined from the limits

$$\frac{\partial J_x}{\partial x} = \frac{1}{A_x} \lim_{\Delta x \rightarrow 0} \frac{Q_x(x_1 + \Delta x, t) - Q_x(x_1, t)}{\Delta x} \quad (1.27)$$

$$\frac{\partial J_y}{\partial y} = \frac{1}{A_y} \lim_{\Delta y \rightarrow 0} \frac{Q_y(y_1 + \Delta y, t) - Q_y(y_1, t)}{\Delta y} \quad (1.28)$$

At any point (x, y) we get

$$\begin{aligned} \frac{\partial c}{\partial t} &= f(x, y, t) - \frac{\partial J_x}{\partial x} - \frac{\partial J_y}{\partial y} \\ &= f(x, y, t) - \nabla \cdot \mathbf{J}. \end{aligned} \quad (1.29)$$

Here ∇ is the del-operator, $\nabla = (\partial/\partial x, \partial/\partial y)$, and $\mathbf{J} = (J_x, J_y)$ is the vector flux. The scalar product $\nabla \cdot \mathbf{J}$ is called the *divergence* of $\mathbf{J}$ and is

$$\nabla \cdot \mathbf{J} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right) \cdot (J_x, J_y) = \frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y}. \quad (1.30)$$

It is seen that if the divergence is positive there exists a net flux out of the point and, vice-versa, if it is negative there is a net flux to the point.

Again, we need to apply Fick's law in order to close the problem. In this case we have c

$$\mathbf{J} = -D \left(\frac{\partial c}{\partial x}, \frac{\partial c}{\partial y} \right) = -D \nabla c. \quad (1.31)$$

Note, ∇c is the gradient of c . In the case where the diffusion coefficient is constant we obtain

$$\begin{aligned} \frac{\partial c}{\partial t} &= f(x, y, t) + \nabla \cdot D \nabla c \\ &= f(x, y, t) + D \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right) \cdot \left(\frac{\partial c}{\partial x}, \frac{\partial c}{\partial y} \right) \\ &= f(x, y, t) + D \left(\frac{\partial}{\partial x} \frac{\partial c}{\partial x} + \frac{\partial}{\partial y} \frac{\partial c}{\partial y} \right) \\ &= f(x, y, t) + D \nabla^2 c \end{aligned} \quad (1.32)$$

The second order differential operator

$$\nabla^2 = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \quad (1.33)$$

is called the *Laplace operator* ².

We return to the bacteria colony. The colony grows radially from a point and the Cartesian coordinates above are rather inconvenient to describe this. Therefore we choose polar coordinates such that

$$c = c(\rho, \theta), \quad (1.34)$$

where ρ is the radial distance to the seeding point and θ the polar angle. Recall that the relation between Cartesian and polar coordinates are

$$x = \rho \cos(\theta) \text{ and } y = \rho \sin(\theta). \quad (1.35)$$

By heavy usage of the chain rule we can use these relations to derive the Laplace operator in polar coordinates

$$\nabla^2 = \left(\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2}{\partial \theta^2} \right). \quad (1.36)$$

A case which will be of interest to us is when the concentration is angle independent, $c = c(\rho, t)$, then the RD-equation is again a one dimensional problem

$$\frac{\partial c}{\partial t} = r(c) + \frac{D}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial c}{\partial \rho} \right) \quad (1.37)$$

if D is constant and if the production function is $r(c)$. We will return to the two dimensional case in Chapter 4. $\diamond$

Question What would be appropriate BCs for the bacteria growth in a Petri dish?

This primer on the RD-equation is not meant as a rigorous mathematical treatment, and we approach the topic in a rather informal and intuitive manner with examples and exercises supported by computer simulations. For a formal treatment, the reader can consult books by Grindrod *The Theory and Applications of Reaction Diffusion Equations: Patterns and Waves* and Smoller *Shock, Waves, and Reactions - Diffusion Equations*.

²and is often (especially among mathematicians) given the symbol Δ

Chapter 2

The steady-state and boundary value problems

An important class of problems deal with systems in the steady-state, where the systems after a time transient no longer change with respect to time. The steady-state we define as

$$\frac{\partial c}{\partial t} = 0, \quad (2.1)$$

and we sometimes highlight the time independence by writing $c = c(x)$. In the steady-state the RD-equation reduces to an ordinary differential equation

$$\frac{d}{dx} \left[D \frac{dc}{dx} \right] + r(c) = 0, \quad (2.2)$$

This equation and the BCs define a boundary value problem and is what we will study in this chapter. In the last optional section we also explore *chemotaxis* which is a phenomenon where cells migrate due to a signaling compound. While this is not a reaction-diffusion phenomenon, we can easily apply what we learn in the first sections and thus further extend the applications.

The boundary value problems we encounter here when r is a linear function of c are examples of the *Sturm-Liouville* problem. The theoretical framework of this general problem is elegant and quite useful, however, we shall not embark on an in-depth treatment here, but simply make a few references.

2.1 Constant coefficients and linear reaction functions

In Example 1 we introduced an RD-equation with a simple linear reaction function, $r = kc$. Let us explore this example, specifically for the case where $k > 0$ and $D > 0$ are constants

$$D \frac{d^2 c}{dx^2} + kc = 0. \quad (2.3)$$

In this first example we apply simple Dirichlet BCs

$$c(0) = c(L) = 0. \quad (2.4)$$

Equation (2.3) is a homogeneous second order linear equation with constant coefficients. The *eigenvalues* for this equation are found from the characteristic equation

$$\lambda^2 + \frac{k}{D} = 0. \quad (2.5)$$

If you need to refresh this, I recommend that you go the exercises for this section. The eigenvalues λ_1 and λ_2 are then

$$\lambda_{1,2} = \pm i \sqrt{\frac{k}{D}}, \quad (2.6)$$

because both coefficients are positive. Since we have distinct eigenvalues the general solution is a linear combination of two exponential functions. Therefore by application of Euler's identity we get

$$\begin{aligned} c(x) &= Ae^{\lambda_1 x} + Be^{\lambda_2 x} = Ae^{i\sqrt{k/D}x} + Be^{-i\sqrt{k/D}x} \\ &= (A + B) \cos\left(\sqrt{k/D}x\right) + i(A - B) \sin\left(\sqrt{k/D}x\right), \end{aligned} \quad (2.7)$$

where A and B are constants we must seek from the BCs.

From the left BC (at $x = 0$) we get

$$c(0) = A + B = 0 \Rightarrow A = -B, \quad (2.8)$$

and using this result together with the right BC (at $x = L$) we have

$$c(L) = i2A \sin\left(\sqrt{k/D}L\right) = 0. \quad (2.9)$$

This relation is of course fulfilled for $A = 0$ and this leads to the *trivial solution* $c = 0$ for all $x \in [0; L]$. Perhaps more interestingly, the equation is also fulfilled when $A \neq 0$ and $\sin(\sqrt{k/D}L) = 0$, that is, for

$$\sqrt{\frac{k}{D}}L = n\pi \text{ where } n \in \mathbb{N}_+. \quad (2.10)$$

Note that since k, D and L are all positive real numbers, the right-hand side must be positive implying that n is a positive integer as we indicate by $\mathbb{N}_+$. Also, notice that for a given set k, D and L we may have $\sqrt{k/D}L \neq n\pi$ for any $n \in \mathbb{N}_+$, hence, in this case only the trivial solution exists.

Assume that Equation(2.10) is fulfilled, then we can substitute the relation into the general solution, Equation (2.7), obtaining

$$c(x) = i2A \sin\left(\frac{n\pi}{L}x\right). \quad (2.11)$$

In our treatment here c is a real valued function and therefore A must be imaginary. Moreover, if we demand $c \geq 0$ (c representing concentration) only $n = 1$ will work defining the length

$$L_0 = \pi \sqrt{\frac{D}{k}} \quad (2.12)$$

constraining the x -coordinate to $0 \leq x \leq L_0$.

We see that even if we have two BCs we cannot determine the constant A (and get a unique solution). What we got from the BCs is the relation between the constants A and B , and the non-trivial solution constraint given in Equation (2.10). We must conclude that there exists infinitely many solutions – if c can take any real value.

The result here, namely, that there does not exist a unique solution may seem surprising if we consider the corresponding experimental bio-chemical system we seek to model. The substance, say, a plankton population features a self-multiplying process, and due to the concentration gradient, the produced plankton is transported via diffusion to the tube-ends where it is removed (how, will not concern us here). Now, if the diffusive transport is sufficiently fast compared to the production, the population vanishes with time (as the plankton is removed at the tube ends), and we simply end in the trivial solution $c = 0$. On the other hand, if diffusion cannot transport the produced plankton to the tube ends fast enough, the concentration will keep increasing leading to c diverging at all interior points; now we must of course evaluate whether our model fulfills its purpose. Finally, there is also a state where there exists a (strict) balance between the diffusive transport out of the system and the production, given by $\sqrt{k/D} = \pi/L_0$, and this allows for a non-trivial solution. In Chapter 4 we revisit these points in more detail.

Let us return to the solutions in Equation (2.11). The *superposition principle for linear systems* states that if Equation (2.11) are solutions to the problem, then a linear combination of these solutions is also a solution. Therefore, we have, in general, the series solution

$$c(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L} x\right), \quad (2.13)$$

where $b_n \in \mathbb{R}$ are constants. This general solution can be understood in a more abstract manner. To this end, first note (or recall) that the sine function can be regarded as a vector. For example, the distributive law holds for the standard addition operator

$$(a + b) \sin\left(\frac{n\pi}{L} x\right) = a \sin\left(\frac{n\pi}{L} x\right) + b \sin\left(\frac{n\pi}{L} x\right) \quad (2.14)$$

where a and b are scalars; the other axioms are just as easily verified. Secondly, the functions $\sin(\pi/L x)$, $\sin(2\pi/L x)$, $\sin(3\pi/L x)$, $\dots$ are *linearly independent* because for any pair $n, m \in \mathbb{N}_+$

$$a \sin\left(\frac{n\pi}{L} x\right) + b \sin\left(\frac{m\pi}{L} x\right) = 0, \quad x \in [0; L], \quad (2.15)$$

implies that $a = b = 0$. Thus, $\sin(\pi/Lx), \sin(2\pi/Lx), \sin(3\pi/Lx), \dots$ span an infinite dimensional vector space, and the solution c is simply a vector itself and represented by the coefficients b_n in the basis spanned by the set

$$\mathcal{V} = \left\{ \sin\left(\frac{n\pi}{L}x\right) : n \in \mathbb{N}_+ \right\}. \quad (2.16)$$

Question Can you verify the statement in Equation (2.13)? (Hint: Show Equation (2.13) is solution to Equation (2.3) with the BCs $c(0) = c(L) = 0$.)

Finally, some terminology: We can rewrite the relation in Equation (2.10) slightly

$$\sqrt{\frac{k}{D}} = \frac{n\pi}{L}. \quad (2.17)$$

Comparing this with Equation (2.5) this motivates us to name $n\pi/L$ an eigenvalue, and we sometimes assign it the symbol λ_n , where n refers to the natural number in the numerator. Note, for this problem we have infinitely many eigenvalues where $0 < \lambda_n < \lambda_m$ if $n < m$. The solution for a given eigenvalue is called an *eigenfunction*; here the sine functions are eigenfunctions. The trivial solution, $c = 0$, is not an eigenfunction. Eigenvalues can, however, be zero!

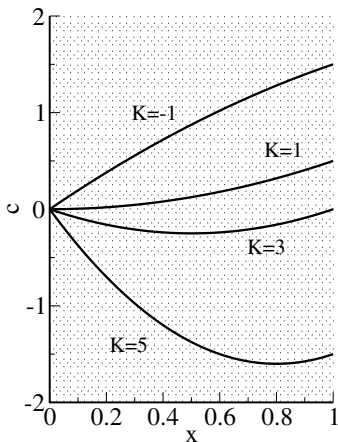
Let us see two more examples.

Example 5 In case the production function is a constant we have an inhomogeneous equation on the form

$$\frac{d^2c}{dx^2} = K, \quad K \in \mathbb{R}. \quad (2.18)$$

We will apply mixed boundaries

$$c(0) = 0 \quad \text{and} \quad \left. \frac{dc}{dx} \right|_{x=L} = 1. \quad (2.19)$$



Clearly, the exponential function cannot be a solution to this problem. Now, let $y = dc/dx$, then $dy/dx = K$, implying that $y = Kx + A$ and therefore

$$c(x) = \frac{K}{2}x^2 + Ax + B. \quad (2.20)$$

From the left BC we get $B = 0$, and since

$$\left. \frac{dc}{dx} \right|_{x=L} = KL + A = 1 \Rightarrow A = 1 - KL \quad (2.21)$$

the solution is

$$c(x) = \frac{K}{2}x^2 + (1 - KL)x. \quad (2.22)$$

This was less painful; in summary, the solution exists and is unique. $\diamond$

Example 6 We return to the simple linear growth problem

$$D \frac{d^2 c}{dx^2} + kc = 0, \quad (2.23)$$

but this time we will apply the BCs

$$c(0) = 1, \quad c(L_0) = 1. \quad (2.24)$$

L_0 is given in Equation (2.12). Now, the general solution is again Equation (2.7). From the first BC we obtain the relation $B = 1 - A$. From the second BC we have

$$c(L_0) = \cos(\pi) + i(2A - 1) \sin(\pi) = -1. \quad (2.25)$$

This conflicts with the right BC which states $c(L_0) = 1$ and is an example of an *ill posed problem*. Both BCs cannot be satisfied and no solution exists. $\diamond$

The examples show that we need to be careful when dealing with boundary value problems. Even if the differential equation looks harmless and the general solution exists it may be an ill posed problem or have infinitely many solutions. We cannot transfer the existence and uniqueness theorems from initial value problems to boundary value problems.

In summary, we have the following possible outcomes:

- There does not exist a solution.
- There exists one unique solution.
- There exist infinitely many solutions.

We will return to existence and uniqueness in the next section.

In this section, we have only dealt with the case where both k and D are constants and independent of position. Clearly, this need not to be the case, e.g., the substance can be distributed in an inhomogeneous medium. These types of problems call for a different approach, which we will treat in Section 2.3.

Exercise 1 Consider the general form for a homogeneous second order differential equation with constant coefficients

$$\frac{d^2 c}{dx^2} + \alpha \frac{dc}{dx} + \beta c = 0 \quad \alpha, \beta \in \mathbb{R}. \quad (2.26)$$

1. Assume $c = c(x)$ has the solution $c(x) = c_0 e^{\lambda x}$. Show that the characteristic equation is

$$\lambda^2 + \alpha \lambda + \beta = 0. \quad (2.27)$$

2. Identify when the characteristic equation has one (repeated) and two distinct solutions.
3. Assume the characteristic equation has two distinct solutions λ_1 and λ_2 . Then both $Ae^{\lambda_1 x}$ and $Be^{\lambda_2 x}$, $A, B \in \mathbb{R}$, are solutions. Verify that the linear combination

$$Ae^{\lambda_1 x} + Be^{\lambda_2 x} \quad (2.28)$$

is also a solution.

Exercise 2 Solve the boundary value problems:

1. $\frac{d^2 c}{dx^2} = K$ with $c(0) = 0$, $c(L) = 0$
2. $\frac{d^2 c}{dx^2} = 0$ with $c(0) = 0$, $c(L) = C_L$
3. $\frac{d^2 c}{dx^2} = -c$ with $\frac{dc}{dx}\big|_{x=0} = 1$, $c(L) = 0$

Relate 1 and 2 to the flow cases in Example 2.

Exercise 3 (Mandatory) Consider the Neumann boundary value problem

$$\frac{d^2 c}{dx^2} + \alpha c = 0 \text{ with } \frac{dc}{dx}\bigg|_{x=0} = \frac{dc}{dx}\bigg|_{x=L} = 0 \quad (2.29)$$

Show that

1. this problem has the trivial solution as only solution when $\alpha < 0$,
2. this problem has infinite many solutions when $\alpha \geq 0$, and
3. that the general solution for $\alpha \geq 0$ is on the form

$$c(x) = \sum_{n=0}^{\infty} a_n \cos\left(\frac{n\pi}{L} x\right), \quad (2.30)$$

where $a_n \in \mathbb{R}$ is a constant.

We shall use these important results later.

Exercise 4 Show that when $k < 0$ and $D > 0$ the trivial solution, $c = 0$, is a solution to the steady-state problem, Equation(2.3). How do you bio-physically argue that this is the only solution?

2.2 Existence and uniqueness of solutions

As illustrated in the previous section, we are not guaranteed a unique or even a single solution to a given boundary value problem. In some cases we can formally prove whether a solution exists and whether it is unique. For example, consider the famous one dimensional *Laplace equation*

$$\frac{d^2 c}{dx^2} = 0. \quad (2.31)$$

Theorem The only solution to the Laplace equation with Dirichlet BCs,

$$c(0) = c(L) = 0, \quad (2.32)$$

is the trivial solution $c = 0$. $\circ$

Before going through the formal proof we can quite easily understand why this is likely true. A good guess for the general solution to the Laplace equation is a simple linear function of the form $c = ax + b$, where a and b are the coefficients we must find from the BCs. The condition $c(0) = 0$ implies that $b = 0$, and only $a = 0$ then satisfies $c(L) = 0$, that is, $c = 0$.

Question Why should we not consider this reasoning as a satisfactory proof for uniqueness?

Proof Assume two solutions exist, namely, φ_1 and φ_2 . We define a third function, φ , as the difference between these two solutions $\varphi = \varphi_1 - \varphi_2$. Importantly, φ fulfills the Laplace equation with the same Dirichlet BCs as the original problem since

$$\frac{d^2 \varphi}{dx^2} = \frac{d^2 \varphi_1}{dx^2} - \frac{d^2 \varphi_2}{dx^2} = 0, \quad (2.33)$$

and $\varphi(0) = \varphi_1(0) - \varphi_2(0) = 0$ and $\varphi(L) = \varphi_1(L) - \varphi_2(L) = 0$. Therefore, if $\varphi = 0$ then $\varphi_1 = \varphi_2$ and the zero solution exists and is unique.

From the product rule we have the following identity

$$\frac{d}{dx} \left(\varphi \frac{d\varphi}{dx} \right) = \left(\frac{d\varphi}{dx} \right)^2 + \varphi \frac{d^2 \varphi}{dx^2} = \left(\frac{d\varphi}{dx} \right)^2, \quad (2.34)$$

where the last relation is due to Equation (2.33). We integrate both sides from 0 to L ; then from the fundamental theorem of calculus the left-hand side becomes

$$\int_0^L \frac{d}{dx} \left(\varphi \frac{d\varphi}{dx} \right) dx = \varphi \frac{d\varphi}{dx} \Big|_{x=0}^{x=L} = 0 \quad (2.35)$$

from the BCs. This implies that the right-hand side is also zero, i.e.,

$$\int_0^L \left(\frac{d\varphi}{dx} \right)^2 dx = 0. \quad (2.36)$$

The integrand is always zero or positive because it is a real function squared, thus, for the integral to be zero we must have $d\varphi/dx = 0$ for $0 \leq x \leq L$. The only way the BCs can be satisfied is then if

$$\varphi = \varphi(0) = \varphi(L) = 0 \text{ for } 0 \leq x \leq L. \quad (2.37)$$

This means that $\varphi_1 = \varphi_2$ (only one solution exists), and since φ is given by the original Laplace equation only the trivial solution is the solution. •

Things change dramatically if we change the BCs.

Theorem The Laplace equation with Neumann BCs

$$\frac{d^2c}{dx^2} = 0, \text{ with } \left. \frac{dc}{dx} \right|_{x=0} = \left. \frac{dc}{dx} \right|_{x=L} = 0, \quad (2.38)$$

does not have a unique solution. ◦

Proof The proof follows the same idea as above and is left to the reader (Come to class if you are stuck!) •

Apart from the exercises below, we will not pursue proving these existence and uniqueness theorems further; again, the interested reader can consult the literature for further details.

Exercise 5 Prove that the only solution to the boundary value problem

$$\frac{d^2c}{dx^2} = c, \quad c(0) = c(L) = 0, \quad (2.39)$$

is the trivial solution.

Exercise 6 Prove that the *Poisson equation* with Dirichlet BCs

$$\frac{d^2c}{dx^2} = f(x), \quad c(0) = c(L) = 0, \quad (2.40)$$

has a unique solution for any continuous function $f : [0; L] \rightarrow \mathbb{R}$.

2.3 Varying coefficients: Series solutions

In Section 2.1 we studied the case where D and k are constants. In general, this is not the case. For example, in a high column of water the biological growth rate coefficient can vary with the depth as the amount of sunlight varies and temperature variations in the system will affect the diffusion coefficient. We sometimes

highlight the position dependency by writing $D = D(x)$ and $k = k(x)$. For varying coefficients the steady-state RD-equation is on the form

$$D(x)\frac{d^2c}{dx^2} + D'(x)\frac{dc}{dx} + k(x)c = 0, \quad (2.41)$$

by application of the product rule and where $D'(x) = dD(x)/dx$. Notice that from our definition of a diffusion process we have $D(x) > 0$. Equation (2.41) is often rewritten on the form

$$\frac{d^2c}{dx^2} + p(x)\frac{dc}{dx} + q(x)c = 0 \quad (2.42)$$

with $p(x) = D'(x)/D(x)$ and $q(x) = k(x)/D(x)$.

In this more general case, the solution for c is not always given by some low order polynomial or a combination of trigonometric functions, and, in fact, the solution may not be expressible in terms of elementary functions. The strategy behind a series solution is to express the solution in terms of its power series representation

$$c(x) = \sum_{n=0}^{\infty} c_n (x - x_0)^n, \quad (2.43)$$

where x_0 is a point typically in the interval $[0; L]$, and c_n are coefficients we need to determine. For this to be meaningful in the first place we must have that the limit of the *the partial sum*

$$s_m(x) = \sum_{n=0}^m c_n (x - x_0)^n \quad (2.44)$$

exists at x as $m \rightarrow \infty$. If this is so, we say the series *converges pointwise* at the point x . A formal definition of pointwise convergence of a function series to a function is based on an $\epsilon - N$ definition:

Definition Let $c : [0; L] \rightarrow \mathbb{R}$, then the series, Equation (2.43), converges pointwise to c at point $x \in [0; L]$ if for all $\epsilon > 0$ there exists an $N(x, \epsilon) \in \mathbb{N}$ such that $|s_m(x) - c(x)| < \epsilon$ for $m \geq N$. $\circ$

A function series can converge pointwise in an interval around x_0 – we say that the series as a non-zero convergence radius. There are different ways to determine the convergence radius; we will here only be using the *ratio test* for power series that states

$$\begin{aligned} &\text{converges if } |x - x_0| \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| < 1 \\ &\text{diverges if } |x - x_0| \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| > 1 \end{aligned}$$

for $c_n \neq 0$. The ratio test is inconclusive when $|x - x_0| \lim_{n \rightarrow \infty} |c_{n+1}/c_n| = 1$. Strictly, the ratio test tests for absolute convergence: if a function series converges absolutely it also converges pointwise.

Example 7 The power series

$$c(x) = \sum_{n=0}^{\infty} 2^n x^n \quad (2.45)$$

converges for $|x| < 1/2$. Notice that $x_0 = 0$ and $c_n = 2^n$. The ratio test gives

$$|x| \lim_{n \rightarrow \infty} \left| \frac{2^{n+1}}{2^n} \right| = |x| \lim_{n \rightarrow \infty} 2 = 2|x|. \quad (2.46)$$

Thus, for $|x| < 1/2$, or $-1/2 < x < 1/2$, the limit fulfills the condition for convergence. The radius of convergence is $1/2$. $\diamond$

Question Will the series in Example 7 converge for $x = 1/2$?

We are now ready for a very powerful result which we will not prove. In our treatment here $D \neq 0$ in any point in the interval $[0; L]$ and all the points in the interval are so-called *ordinary points*. If a point is not an ordinary point it is a *singular point*, and this occurs for points where $D = 0$. The functions p and q are not defined in singular points. Now, if x is an ordinary point we are guaranteed a general solution on the form of Equation (2.43) in some neighbourhood around the point x . That is, the function c can be written as power series with a convergence radius larger than zero. Moreover, if a function is given by a power series in some interval around x , the derivative of the function in x is given by the sum of the term wise derivatives of the series.

The idea is simple: if p and q are sufficiently smooth we should expect that the general solution to the differential equation exists and, furthermore, that the solution can, locally, be represented by a power series. Of course, we may still have an ill posed boundary value problem prohibiting a particular solution.

Example 8 To illustrate the method, we revisit a simple problem we already know, namely, the special case where diffusion and growth rate are constants, see Section 2.1. Later we let D and k be functions of x . Recall, the steady-state RD-equation reads

$$D \frac{d^2 c}{dx^2} + kc = 0, \quad (2.47)$$

or equivalently

$$\frac{d^2 c}{dx^2} + \lambda c = 0, \quad (2.48)$$

where $\lambda = k/D > 0$ since both rate and diffusion coefficients are positive. First, we notice that in the general form we get $p = 0$ and $q = \lambda$ so any point is an ordinary point. We will choose $x_0 = 0$, hence, we look for a series solution on the form

$$c(x) = \sum_{n=0}^{\infty} c_n x^n. \quad (2.49)$$

We may now substitute Equation (2.49) into Equation (2.47) and compare terms in order to extract the coefficients c_n . However, the problem involves a second order derivative which we must deal with; this is an ordinary problem and we can simply differentiate term wise. Differentiating once

$$\frac{dc}{dx} = c_1 + 2c_2x + 3c_3x^2 + 4c_4x^3 + \dots = \sum_{n=1}^{\infty} nc_nx^{n-1} \quad (2.50)$$

and twice

$$\frac{d^2c}{dx^2} = 2c_2 + 6c_3x + 12c_4x^2 + \dots = \sum_{n=2}^{\infty} n(n-1)c_nx^{n-2}. \quad (2.51)$$

Before substitution it is often a practical convenience if the starting indices are the same; we can achieve this by defining a shifting index, here denoted m . For the 1st order derivative we let $m = n - 1$ and therefore $n = m + 1$, and Equation (2.50) is written as

$$\frac{dc}{dx} = \sum_{n=1}^{\infty} nc_nx^{n-1} = \sum_{m=0}^{\infty} (m+1)c_{m+1}x^m. \quad (2.52)$$

For the 2nd order derivative we let $m = n - 2$. Then $n = m + 2$ and

$$\frac{d^2c}{dx^2} = \sum_{n=2}^{\infty} n(n-1)c_nx^{n-2} = \sum_{m=0}^{\infty} (m+2)(m+1)c_{m+2}x^m. \quad (2.53)$$

The index symbol is, of course, changeable so we can simply rename m back to n . Substitution of Equations (2.49) and (2.53) into Equation (2.48) we obtain

$$\sum_{n=0}^{\infty} (n+2)(n+1)c_{n+2}x^n = -\lambda \sum_{n=0}^{\infty} c_nx^n. \quad (2.54)$$

This relation is true if each power term in the two series are equal, that is, if

$$(n+2)(n+1)c_{n+2} = -\lambda c_n \implies c_{n+2} = -\frac{\lambda c_n}{(n+2)(n+1)}. \quad (2.55)$$

This is the *recurrence relation*. From the recurrence relation we can determine all coefficients if we know c_0 and c_1

$$c_2 = -\frac{\lambda c_0}{1 \cdot 2}, \quad c_3 = -\frac{\lambda c_1}{2 \cdot 3}, \quad c_4 = -\frac{\lambda c_2}{3 \cdot 4} = \frac{\lambda^2 c_0}{1 \cdot 2 \cdot 3 \cdot 4}, \quad c_5 = -\frac{\lambda c_3}{4 \cdot 5} = \frac{\lambda^2 c_1}{2 \cdot 3 \cdot 4 \cdot 5}$$

and so on. Notice that c_0 appears in the terms where n is even, and c_1 in the terms where n is odd. We can then write the general solution on the form of an even series and an odd series

$$\begin{aligned} c(x) &= c_0 \left(1 - \frac{\lambda}{2!}x^2 + \frac{\lambda^2}{4!}x^4 - \dots \right) + c_1 \left(x - \frac{\lambda}{3!}x^3 + \frac{\lambda^2}{5!}x^5 - \dots \right) \\ &= c_0 \sum_{n=0}^{\infty} \frac{(-1)^n \lambda^n}{2n!} x^{2n} + c_1 \sum_{n=0}^{\infty} \frac{(-1)^n \lambda^n}{(2n+1)!} x^{2n+1}. \end{aligned} \quad (2.56)$$

This is just a linear combination of the Taylor series for cosine and sine

$$\begin{aligned}\frac{\cos(\sqrt{\lambda}x)}{\sqrt{\lambda}} &= \sum_{n=0}^{\infty} \frac{(-1)^n \lambda^n}{2n!} x^{2n} \\ \frac{\sin(\sqrt{\lambda}x)}{\sqrt{\lambda}} &= \sum_{n=0}^{\infty} \frac{(-1)^n \lambda^n}{(2n+1)!} x^{2n+1}\end{aligned}\quad (2.57)$$

so we arrive at the final form

$$c(x) = \frac{c_0}{\sqrt{\lambda}} \cos(\sqrt{\lambda}x) + \frac{c_1}{\sqrt{\lambda}} \sin(\sqrt{\lambda}x). \quad (2.58)$$

This is the same result we obtained in Equation (2.7); which we should also expect, of course. The coefficients can, be determined by BCs unless the problem is ill-posed. $\diamond$

The fact that the solution power series is the Taylor series is a general result. So, even if the solution is an unknown function we can derive the function's Taylor series (and give it a name if we wish).

Example 9 We consider the case where the diffusion coefficient is constant, $D(x) = D$ and where the growth rate coefficient is linear with respect to space $k(x) = k_0 x$. We have

$$D \frac{d^2 c}{dx^2} + k_0 x c = 0 \implies \frac{d^2 c}{dx^2} = -\lambda x c \quad (2.59)$$

where $\lambda = k_0/D$. In this example, we consider the BCs

$$c(0) = 0 \text{ and } c(1) = 1. \quad (2.60)$$

We have that $p = 0$ and $q = \lambda x$, so any point is an ordinary point. Again, let $x_0 = 0$ such that

$$c(x) = \sum_{n=0}^{\infty} c_n x^n. \quad (2.61)$$

Now, we substitute the series Equation. (2.61) and (2.53) - with index symbol n - into Equation(2.59) we arrive at

$$\sum_{n=0}^{\infty} (n+2)(n+1) c_{n+2} x^n + \lambda x \sum_{n=0}^{\infty} c_n x^n = 0 \quad (2.62)$$

or by distributing x in the last term

$$\sum_{n=0}^{\infty} (n+2)(n+1) c_{n+2} x^n + \lambda \sum_{n=0}^{\infty} c_n x^{n+1} = 0. \quad (2.63)$$

It is a bit harder to derive the recurrence relation for this case. It may be deduced by writing out a few terms

$$\begin{aligned}
x^0 &: 2c_2 = 0 \implies c_2 = 0 \\
x^1 &: 3 \cdot 2c_3 + \lambda c_0 = 0 \implies c_3 = -\lambda c_0 / (3 \cdot 2) \\
x^2 &: 4 \cdot 3c_4 + \lambda c_1 = 0 \implies c_4 = -\lambda c_1 / (4 \cdot 3) \\
x^3 &: 5 \cdot 4c_5 + \lambda c_2 = 0 \implies c_5 = 0 \\
x^4 &: 6 \cdot 5c_6 + \lambda c_3 = 0 \implies c_6 = -\lambda c_3 / (6 \cdot 5) \\
x^5 &: 7 \cdot 6c_7 + \lambda c_4 = 0 \implies c_7 = -\lambda c_4 / (7 \cdot 6) \\
&\vdots
\end{aligned}$$

Clearly, there is a structure here

$$c_2 = 0 \text{ and } c_n = -\frac{\lambda c_{n-3}}{n(n-1)} \text{ for } n \geq 3. \quad (2.64)$$

Notice that since $c_2 = 0$ we have that $c_5 = c_8 = c_{11} = \dots = 0$. Moreover, from the recurrence relation we get

$$\begin{aligned}
c_6 &= -\frac{\lambda c_3}{6 \cdot 5} = \frac{\lambda^2 c_0}{6 \cdot 5 \cdot 3 \cdot 2} \\
c_7 &= -\frac{\lambda c_4}{7 \cdot 6} = \frac{\lambda^2 c_1}{7 \cdot 6 \cdot 4 \cdot 3}
\end{aligned}$$

and

$$\begin{aligned}
c_9 &= -\frac{\lambda c_6}{9 \cdot 8} = -\frac{\lambda^3 c_0}{9 \cdot 8 \cdot 6 \cdot 5 \cdot 3 \cdot 2} \\
c_{10} &= -\frac{\lambda c_7}{10 \cdot 9} = -\frac{\lambda^3 c_1}{10 \cdot 9 \cdot 7 \cdot 6 \cdot 4 \cdot 3}
\end{aligned}$$

and so forth. This leaves only two unknowns, namely, c_0 and c_1 , and we again write the general solution as a linear combination of two series

$$\begin{aligned}
c(x) &= c_0 \left(1 - \frac{\lambda}{3 \cdot 2} x^3 + \frac{\lambda^2}{6 \cdot 5 \cdot 3 \cdot 2} x^6 - \frac{\lambda^3}{9 \cdot 8 \cdot 6 \cdot 5 \cdot 3 \cdot 2} x^9 + \dots \right) \\
&+ c_1 \left(x - \frac{\lambda}{4 \cdot 3} x^4 + \frac{\lambda^2}{7 \cdot 6 \cdot 4 \cdot 3} x^7 - \frac{\lambda^3}{10 \cdot 9 \cdot 7 \cdot 6 \cdot 4 \cdot 3} x^{10} + \dots \right) \quad (2.65)
\end{aligned}$$

In practice we cannot add up infinite terms, and some times we may not even be able to find a general recursion relation. We therefore often use the partial sum

$$s_m(x) = \sum_{n=0}^m c_n x^n. \quad (2.66)$$

Applying the left boundary condition, $c(0) = 0$, we get $c_0 = 0$ no matter the choice of truncation and then up to, say 10th order, we have

$$c(x) \approx c_1 s_{10}(x) = c_1 \left(x - \frac{\lambda}{4 \cdot 3} x^4 + \frac{\lambda^2}{7 \cdot 6 \cdot 4 \cdot 3} x^7 - \frac{\lambda^3}{10 \cdot 9 \cdot 7 \cdot 6 \cdot 4 \cdot 3} x^{10} \right) \quad (2.67)$$

Here we determine c_1 from the partial sum: since $c(1) \approx c_1 s_{10}(1) = 1$ we get $c_1 = 1/s_{10}(1)$. Different partial sums are plotted for the case $\lambda = 10$. Notice that already after three terms, or 7th order in x , we see a good approximation. $\diamond$

Proposition: The power series solution in Equation (2.65) converges for all $x \in \mathbb{R}$.

To show this, we write the terms in Equation (2.65) as

$$\frac{(-1)^n x^{3n+1}}{(3n+1) \cdot (3n) \dots 4 \cdot 3} \quad n = 1, 2, \dots \quad (2.68)$$

Recall $c_0 = 0$. It is understood that for $n = 1$ we include one multiplication in the denominator, $n = 2$ two multiplications, etc. For index n we have the coefficient

$$c_n = \frac{(-1)^n}{(3n+1) \cdot (3n) \dots 4 \cdot 3} \quad (2.69)$$

and for index $n + 1$

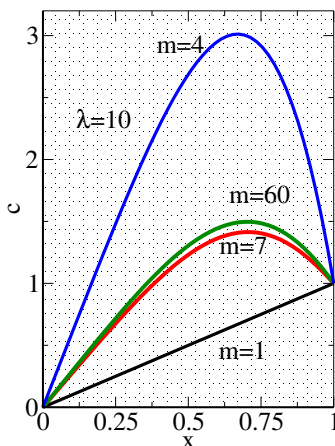
$$c_{n+1} = \frac{(-1)^{n+1}}{(3(n+1)+1) \cdot (3(n+1)) \cdot (3n+1) \cdot (3n) \dots 4 \cdot 3} \quad (2.70)$$

This implies that

$$|x| \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| = |x| \lim_{n \rightarrow \infty} \left| \frac{1}{(3(n+1)+1) \cdot (3(n+1))} \right| = 0. \quad (2.71)$$

This limit is independent of x and the series solution converges on $\mathbb{R}$. This implies that the power series solution found in Equation (2.65) is the solution for all x .

Partial sums with $c_1 = 1/s_m(x)$



Exercise 7 Determine the convergence radius for the following function series using the ratio test

- | | |
|--|--|
| a) $\sum_{n=0}^{\infty} x^n$ (Geometric series) | b) $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ |
| c) $\sum_{n=0}^{\infty} (n!) x^n$ | d) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n} (x-1)^n$ |
| e) $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$ | f) $\sum_{n=0}^{\infty} \frac{(x-x_0)^n}{a^n (n+1)}$ where $a > 1$ |

Exercise 8 For the following three cases below: (i) identify the functions p and q and show that $x_0 = 0$ is an ordinary point, (ii) derive the recurrence relation (if possible), and (iii) find the general power series solution (you may write this in terms up to 10th order with respect to x).

1. $D(x) = D_0(1 - x)$ and k constant
2. $D(x) = D_0(1 - x)$ and $k(x) = k_0x$
3. $D(x) = D_0(1 - x^2)$ and k constant

Exercise 9 Find the particular solution for the exercise above using BCs

1. $c(0) = 0$ and $c(1) = 1$
2. $\frac{dc}{dx}\big|_{x=0} = 0$ and $\frac{dc}{dx}\big|_{x=1} = 0$
3. $\frac{dc}{dx}\big|_{x=0} = 1$ and $\frac{dc}{dx}\big|_{x=1} = -1$

Plot the solutions (where relevant) using partial sums with $m = 1, 5, 10$. Find (where relevant) the radius of convergence.

Exercise 10 Find the series solution for the problem in Example 9 using $x_0 = 1$. (Hints: You may want to use the relation $x = 1 + x - 1$ and the variable transformation $z = x - 1$.)

Exercise 11 Series solutions may be applied to first order differential equations, initial value problems, non-linear problems, etc. Use a series solution around $x_0 = 0$ to solve the following boundary value problem where the coefficient is constant

$$\frac{d^2c}{dx^2} + K = 0, \quad K \in \mathbb{R} \quad (2.72)$$

with

$$c(0) = 0 \text{ and } \frac{dc}{dx}\bigg|_{x=L} = 1. \quad (2.73)$$

Compare your result with Example 5.

Exercise 12 Airy's equation is used in modelling waves, optics, and more. It reads

$$\frac{d^2c}{dx^2} - xc = 0 \quad \text{where} \quad -\infty < x < \infty \quad (2.74)$$

Let $x_0 = 0$ and find the solution for $c(-2) = 0$ and $c(0) = 1$. Plot the partial sum for $m = 1, 10, 100, 1000$ in the interval $-10 < x < 2$.

2.4 Non-linear reaction functions

Bio-chemical reactions are often non-linear. The first way to approach such a situation is, almost always, to study the system in some special cases or to do a linear analysis in the hope that this gives some qualitative understanding of the system. Sometimes, however, we wish to extend our analysis, e.g., if we want a better quantitative output from the model.

Consider the boundary value problem

$$D \frac{d^2 c}{dx^2} + kc^2 = 0 \quad \text{with } c(0) = 0 \quad \text{and } c(1) = 1. \quad (2.75)$$

Notice that the reaction term is non-linear and we cannot use the standard techniques discussed so far to solve this. We can re-arrange this a bit by dividing with D , and introduce a new parameter $\epsilon = k/D$, $D \neq 0$, such that the dynamical equation becomes

$$\frac{d^2 c}{dx^2} + \epsilon c^2 = 0. \quad (2.76)$$

In the case where the diffusion coefficient is large compared to the rate coefficient, $D \gg k$, we have that ϵ is small. Indeed, we will expect that the solution in this case is “close” to the solution for $d^2 c/dx^2 = 0$, where there is no reaction process. We can think of ϵ as a parameter that controls a (small) perturbation of the system away from the pure diffusion case. For this reason ϵ is called the *perturbation parameter* and Equation (2.76) forms a *regular perturbation problem*. A problem is called regular if setting $\epsilon = 0$ does not change the order or basic character of the governing equation. A non-regular problem is illustrated later.

Once we have defined a small system parameter and have a regular perturbation problem, we can attempt to do an analysis based on the *regular perturbation method*. This method is based on the fundamental assumption (called an *ansatz*) that the function c can be represented by power series with respect to the perturbation parameter. At first, this does not appear to be very helpful since we are interested in c as a function of the spatial coordinate x and not the parameter ϵ , however, the power series will result in a, some times, tractable structure. The regular perturbation method involves the following five steps:

1. Assume that a solution $c = c(x)$ exists. Furthermore, assume that c can be written in terms of a power series with respect to ϵ

$$c(x) = c_0(x) + \epsilon c_1(x) + \epsilon^2 c_2(x) + \epsilon^3 c_3(x) + \dots \quad (2.77)$$

where $c_0, c_1, \dots$ are functions of x and independent of ϵ . We now hope that c is “well-approximated” with only a few terms, that is, we hope that the function series converges to the actual function quickly.

2. Insert the power series for c into the dynamical equation for c , Equation (2.76),

$$\frac{d^2}{dx^2}(c_0 + \epsilon c_1 + \epsilon^2 c_2 + \dots) + \epsilon(c_0 + \epsilon c_1 + \epsilon^2 c_2 + \dots)^2 = 0. \quad (2.78)$$

3. Collect term-wise with respect to powers of ϵ . To fulfill the equation we equate all terms with zero; thus, collecting terms with equal powers of ϵ we obtain

$$\epsilon^0 : \quad \frac{d^2 c_0}{dx^2} = 0 \quad (2.79)$$

$$\epsilon^1 : \quad \frac{d^2 c_1}{dx^2} + c_0^2 = 0 \quad (2.80)$$

$$\epsilon^2 : \quad \frac{d^2 c_2}{dx^2} + 2c_0 c_1 = 0 \quad (2.81)$$

$$\vdots$$

We see that this results in a set of tractable problems (albeit in general quite tedious). From Equation (2.79) we obtain c_0 (once the appropriate BCs are specified). This result is substituted into Equation (2.80) and c_1 can be found (once the appropriate BCs are specified), and so forth. Notice that in this case each problem is linear.

4. Solve for $c_0, c_1, \dots$

Here we first need to define the BCs. We make a simple choice, namely,

$$c_0(0) = 0 \text{ and } c_0(1) = 1. \quad (2.82)$$

Now since

$$c(0) = c_0(0) + \epsilon c_1(0) + \epsilon^2 c_2(0) + \dots = 0 \quad (2.83)$$

and

$$c(1) = c_0(1) + \epsilon c_1(1) + \epsilon^2 c_2(1) + \dots = c_0(1) = 1 \quad (2.84)$$

this implies that the BCs for $n = 1, 2, 3 \dots$ are $c_n(0) = 0$ and $c_n(1) = 0$. Notice, the BCs can be chosen differently. Then, solving

$$\epsilon^0 : \quad \frac{d^2 c_0}{dx^2} = 0, \quad c_0(0) = 0, c_0(1) = 1 \Rightarrow c_0(x) = x \quad (2.85)$$

$$\epsilon^1 : \quad \frac{d^2 c_1}{dx^2} + x^2 = 0, \quad c_1(0) = 0, c_1(1) = 0 \Rightarrow c_1(x) = \frac{x - x^4}{12} \quad (2.86)$$

$$\vdots$$

5. Collect the terms in accordance with Equation (2.77), here to first order in ϵ

$$c(x) = x + \frac{\epsilon}{12}(x - x^4) + \dots \quad (2.87)$$

This is the solution from the regular perturbation method; we here stop at first order already, but one can continue (see the exercises).

In this example, the first order term, a measure of the effects from the reaction, has a pre-factor of $\epsilon/12$. Therefore, the reaction process has only a small effect on the concentration profile even if ϵ is much larger than zero (and we move away from the regime $D \gg k$).

The truncated power series above is formally known as the *Poincaré expansion* and generally as an *asymptotic expansion*, and the approximation is written as

$$c(x) \sim \sum_{n=0}^N \epsilon^n c_n(x). \quad (2.88)$$

The *remainder* can now be defined as

$$R_N(x, \epsilon) = c(x) - \sum_{n=0}^N \epsilon^n c_n(x) \quad (2.89)$$

and is a measure for the difference between the actual solution and the Poincaré expansion. Often we do not have an analytical expression for c itself, but have to use a numerical computation. We can now define what we mean by the term well-approximated that we used above: The Poincaré expansion (with N terms) is an asymptotic approximation to c if for all $M \leq N$ and $x \in [0; L]$ we have

$$\lim_{\epsilon \rightarrow 0} \frac{R_M(x, \epsilon)}{\epsilon^M c_M(x)} = 0 \text{ for any fixed } x \in [0; L]. \quad (2.90)$$

In words: We want the remainder to go to zero as ϵ goes to zero and faster than the last term in the expansion. The concept of an asymptotic approximation is perhaps best illustrated with an example where the solution is known.

Example 10 The perturbation method is applicable for different types of non-linear problems, not only differential equations. Consider the quadratic problem (example from Hinc *Perturbation Methods*)

$$c^2 - \epsilon c - 1 = 0. \quad (2.91)$$

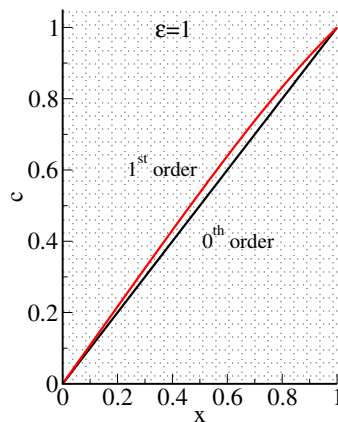
Proposition: The Poincaré expansion

$$c \sim c_0 + \epsilon c_1 + \epsilon^2 c_2 \quad (2.92)$$

is an asymptotic approximation to the quadratic problem, Equation (2.91).

First, the solution is found from the quadratic formula

$$c = \frac{1}{2} \left(\epsilon \pm \sqrt{\epsilon^2 + 4} \right). \quad (2.93)$$



Notice that for $\epsilon = 0$ we have $c = \pm 1$. Substituting the expansion and collecting the terms we have up to second order

$$\epsilon^0 : c_0^2 - 1 = 0 \Rightarrow c_0 = \pm 1 \quad (2.94)$$

$$\epsilon^1 : 2c_0c_1 - c_0 = 0 \Rightarrow \begin{cases} 2c_1 - 1 = 0 \\ -2c_1 + 1 = 0 \end{cases} \Rightarrow c_1 = \frac{1}{2} \quad (2.95)$$

$$\epsilon^2 : 2c_0c_2 + c_1^2 - c_1 = 0 \Rightarrow \begin{cases} 2c_2 - \frac{1}{4} = 0 \\ 2c_2 + \frac{1}{4} = 0 \end{cases} \Rightarrow c_2 = \pm \frac{1}{8} \quad (2.96)$$

hence, the roots to second order in perturbation parameter are

$$\pm 1 + \frac{\epsilon}{2} \pm \frac{\epsilon^2}{8}. \quad (2.97)$$

We have to show the limits in Equation (2.90) (noting that x is not in the current problem).

For $M = 0$ we have $c_0 = \pm 1$ and

$$R_0(\epsilon) = \frac{1}{2}(\epsilon \pm \sqrt{\epsilon^2 + 4}) \mp 1 \quad (2.98)$$

quickly see that

$$\lim_{\epsilon \rightarrow 0} \frac{R_0(\epsilon)}{c_0} = \lim_{\epsilon \rightarrow 0} \frac{1}{2}(\epsilon \mp \sqrt{\epsilon^2 + 4}) \pm 1 = 0. \quad (2.99)$$

For $M = 1$

$$\lim_{\epsilon \rightarrow 0} \frac{R_1(\epsilon)}{\epsilon c_1} = \lim_{\epsilon \rightarrow 0} \frac{\pm \sqrt{\epsilon^2 + 4} \mp 2}{\epsilon} = \lim_{\epsilon \rightarrow 0} \frac{\epsilon}{\sqrt{\epsilon^2 + 4}} = 0. \quad (2.100)$$

The last limit is found using L'Hopital's rule. Finally, for $M = 2$ we have

$$\lim_{\epsilon \rightarrow 0} \frac{R_2(\epsilon)}{\epsilon^2 c_2} = \lim_{\epsilon \rightarrow 0} \frac{\pm 4\sqrt{\epsilon^2 + 4} \mp 8 \mp \epsilon^2}{\epsilon^2} = 0, \quad (2.101)$$

applying L'Hopital's rule twice. Thus, the Poincaré expansion with $N = 2$ is an asymptotic approximation to the quadratic equation, Equation (2.91). $\diamond$

Be aware, the perturbation method may not always be helpful, at least not without performing additional tricks. Consider a slightly different quadratic equation from above

$$\epsilon c^2 - c - 1 = 0. \quad (2.102)$$

(Again, an example from Hince.) We immediately see that when $\epsilon = 0$ there is only one solution, but for $\epsilon \neq 0$ there are two. A problem where the solution shows such qualitative change setting $\epsilon = 0$ is called a *singular perturbation problem*. Equation (2.91) above is a regular perturbation problem since the number of solutions did not change by varying ϵ around zero. Performing the same steps as we did for regular problems when we deal with singular problems, we end up with the expansion

$$c = 1 - \epsilon + 2\epsilon^2 + \dots \quad (2.103)$$

Compare this to the quadratic formula

$$c = \frac{1}{2\epsilon}(1 \pm \sqrt{1 - 4\epsilon}) \quad \epsilon \neq 0. \quad (2.104)$$

These are two very different results, and since we trust the latter we must disregard the simple Poincaré expansion.

Singular perturbation problems are not just far-fetched curiosities. Consider the steady-state RD-equation in the reaction dominated regime, $k \gg D$. Here we can introduce the perturbation parameter $\epsilon = D/k$ and we can write

$$\epsilon \frac{d^2 c}{dx^2} + c^2 = 0. \quad (2.105)$$

As $\epsilon \rightarrow 0$ the diffusion term vanishes and the problem is algebraic only allowing $c = 0$ for $0 < x < L$; the system qualitative behavior changes dramatically as we approach the limit $\epsilon \rightarrow 0$. This is a singular perturbation problem. We will later encounter such problem, and the naive outer solution will, in fact, be very helpful here.

Exercise 13 Find the second order perturbation term for the boundary value problem Equation (2.76). Plot c_0 , c_1 , c_2 as functions of x .

Exercise 14 Taylor expand to second order the exact solution to the quadratic problem, Equation (2.93), in terms of ϵ around $\epsilon = 0$. Compare with the perturbation solution.

Exercise 15 Consider the initial value problem

$$\frac{dc}{dx} - \epsilon c^2 = 0, \quad c(1) = 1 \quad (2.106)$$

Find the analytical solution. Then use regular perturbation method up to second order in ϵ to approximate the solution. Is the Poincaré expansion with $m = 1$ an asymptotic approximation?

Exercise 16 Consider the boundary value problem

$$\frac{d^2 c}{dx^2} + c - \epsilon c^2 = 0, \quad c(0) = c(1) = 1 \quad (2.107)$$

Use regular perturbation method up to second order and find the approximate solution...or die trying.

Exploration 1 Figure 2.1 illustrates a cylindrical pore embedded in a membrane. The pore allows for transport of a chemical signaling compound C. In the steady-state, C has concentration C_0 on the “outside” and C_1 on the “inside”. Careful

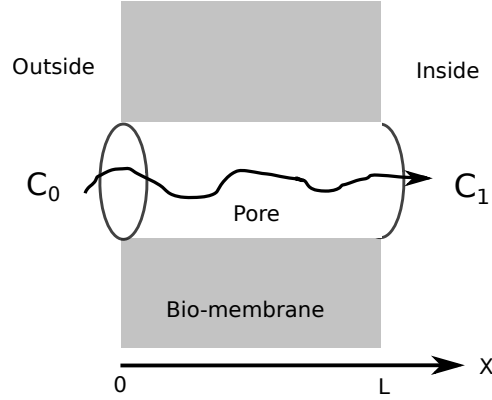


Figure 2.1: Illustration of the bio-pore.

measurements have established that $C_1 < C_0$, and from this we expect that a reaction inside the pore consumes the chemical C .

Let the steady-state concentration of C in the pore be denoted $c = c(x)$ where $0 \leq x \leq L$. The dynamics is modelled from the steady-state RD-equation with mixed BCs

$$r(c) + D \frac{d^2 c}{dx^2} = 0, \quad c(0) = C_0 \quad \text{and} \quad \left. \frac{dc}{dx} \right|_{x=L} = 0. \quad (2.108)$$

1. Show that the condition $C_1 < C_0$ cannot be met if the reaction term in Equation (2.108) is omitted.
2. Show that if $r(c) = -k_1 c$ the concentration profile is given by

$$c(x) = C_0 \left[e^{-\alpha L} \frac{\sinh(\alpha x)}{\cosh(\alpha L)} + e^{-\alpha x} \right], \quad (2.109)$$

where $\alpha = \sqrt{k_1/D}$

3. Now let $r(c) = -k_2 c^2$. Find the solution up to first order in the perturbation parameter $\varepsilon = k_2/D$. Explain the fundamental physical unit that determines the validity of this approximate solution.

2.5 Chemotaxis

Experimentally it has been observed that some cells, e.g., E-coli, can register a chemical concentration gradient and move either down or up along this gradient. The chemical compound can be a hormone and cell motion is realised by, for example, flagella propulsion. The phenomenon is called *chemotaxis*. To model chemotaxis we need to change the fluxes in the system, and if diffusion and chemotaxis are

independent processes we can write the flux with two terms: (i) a term describing the usual diffusive flux and (ii) a term describing the flux due to chemotaxis.

Consider a cell population with concentration $c_1 = c_1(x, t)$ mixed with a chemical compound having concentration $c_2 = c_2(x, t)$. The *Keller-Segel model* for the flux of the cells reads

$$J_1 = \chi c_1 \frac{\partial c_2}{\partial x} - D \frac{\partial c_1}{\partial x}, \quad (2.110)$$

where χ is the *chemotactic strength* and D the usual diffusion coefficient for the cells. What the two terms describe should be clear. If the chemotactic strength or the gradient of the chemical is zero, then, of course, no flux occurs due to chemotaxis. The product in the chemotaxis term between c_1 and $\partial c_2 / \partial x$ give raise to a (potential) non-linearity, and ensures that there is no flux of cells at point when there are no cells even if the gradient of the chemical is non-zero.

Substitution into the balance equation for c_1 gives

$$\frac{\partial c_1}{\partial t} = r(c_1) - \chi \frac{\partial}{\partial x} \left(c_1 \frac{\partial c_2}{\partial x} \right) + D \frac{\partial^2 c_1}{\partial x^2} \quad (2.111)$$

if χ and D are independent of spatial coordinate x . In the steady-state this is simply

$$r(c_1) - \chi \frac{d}{dx} \left(c_1 \frac{dc_2}{dx} \right) + D \frac{d^2 c_1}{dx^2} = 0. \quad (2.112)$$

Question Will the cells move up or down the chemical gradient if $\chi > 0$? What if $\chi < 0$?

Example 11 Assume that we somehow can control the local concentration of the chemical compound. For example, let

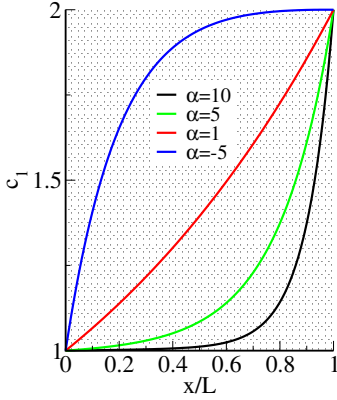
$$c_2 = Kx/L \text{ such that } c_2(0) = 0 \text{ and } c_2(L) = K. \quad (2.113)$$

Let the cells feature a linear growth rate, $r = kc_1$, and use Dirichlet BCs $c_1(0) = C_0$ and $c_1(L) = 2C_0$, where $C_0 \neq 0$. The equation (for the steady-state) is obtained using the product rule

$$kc_1 - \chi \frac{d}{dx} \left(c_1 \frac{dc_2}{dx} \right) + D \frac{d^2 c_1}{dx^2} = 0 \Rightarrow \frac{k}{D} c_1 - \frac{\chi K}{DL} \frac{dc_1}{dx} + \frac{d^2 c_1}{dx^2} = 0. \quad (2.114)$$

Define the parameter $\alpha = \chi K / (DL)$ then the characteristic polynomial is

$$\lambda^2 - \alpha \lambda + \frac{k}{D} = 0, \quad (2.115)$$



implying that

$$\lambda_{1,2} = \frac{1}{2} \left(\alpha \pm \sqrt{\alpha^2 - 4k/D} \right). \quad (2.116)$$

For real eigenvalues, $\alpha^2 > 4k/D$, we then have the general solution

$$c_1(x) = Ae^{\lambda_1 x} + Be^{\lambda_2 x}. \quad (2.117)$$

Applying the BCs we obtain

$$c_1(x) = C_0 \left[e^{\lambda_1 x} + \frac{2 - e^{\lambda_1 L}}{e^{\lambda_2 L} - e^{\lambda_1 L}} (e^{\lambda_2 x} - e^{\lambda_1 x}) \right]. \quad (2.118)$$

Note, by introducing chemotaxis the concentration profile can change from a concave function to a convex function depending on the sign of α , that is, on the sign of the chemotactic strength, χ . $\diamond$

In the general case, we may not be able to control the concentration of the chemical compound and we then need a dynamical equation for c_2 as well. Equation (2.113) is equivalent to the case, where the chemical is not consumed by the cells and simply diffuses according to Fick's law, that is, c_2 follows the Laplace equation

$$\frac{d^2 c_2}{dx^2} = 0 \text{ with } c_2(0) = \text{ and } c_2(L) = K. \quad (2.119)$$

We will not pursue the general case here.

Exercise 17 Solve the chemotaxis problem in Example 11 for the situation where the reaction function is zero everywhere, $r = 0$.

Exercise 18 Let $c_2 = kx^2$. Show that the steady-state chemotaxis model can be written on the form

$$D \frac{d^2 c_1}{dx^2} + \alpha x \frac{dc_1}{dx} + f(c_1) = 0. \quad (2.120)$$

Let $f(c_1) = kc_1$ and find the general solution.

Chapter 3

The diffusion equation

If the system is non-reactive, $r = 0$, the RD-equation reduces to the diffusion equation. If the diffusion coefficient is constant this is in one dimension and in Cartesian coordinates

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2}. \quad (3.1)$$

To solve this problem we need an IC, $f(x) = c(x, 0)$, and BCs which can be of the Dirichlet type, Neuman type or a mix of the two.

Before we explore the diffusion equation, we must first acquire some knowledge about the famous Fourier series and the Euler-Fourier equations as they play a key role in the exploration of the diffusion equation.

3.1 Periodic functions and the Fourier series

In one dimension, a periodic function is a function defined on a subset S in $\mathbb{R}$ and which has repeating values such that for $T \neq 0$

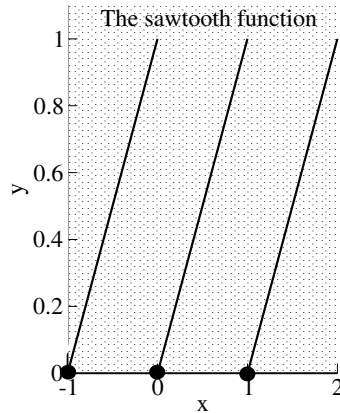
$$f(x) = f(x + T) \quad (3.2)$$

where $x \in S$ and $x+T \in S$. T is called the function period. T is not unique; the least value for T is referred to as the *fundamental period*, e.g., the functions cosine and sine have fundamental period 2π .

Another example is the *sawtooth function*

$$\text{saw}(x) = x - \lfloor x \rfloor, \quad (3.3)$$

here $\lfloor x \rfloor$ is the floor function, defined such that $\lfloor -0.5 \rfloor = -1$, $\lfloor 0.5 \rfloor = 0$, $\lfloor 1 \rfloor = 1$, and so forth. The sawtooth function has a fundamental period of 1, and since the left and right function limits in points



$x = \dots - 2, -1, 0, 1, 2, \dots$ exist, but are different the function features *jump discontinuities* in these points.

Another way of defining the sawtooth function is to *periodically expand* the function $f(x) = x$, where $0 \leq x < 1$. The expansion is given as

$$\text{saw}(x) = \text{saw}(x + n) , \quad (3.4)$$

with $n \in \mathbb{Z}$.

The sawtooth function is defined on the entire real line and we have $S = \mathbb{R}$. We can define another (and related) periodic function with fundamental period 1 using the periodic expansion of $f(x) = x$, but where $0 < x < 1$ such that $S = \mathbb{R} \setminus \{\mathbb{Z}\}$.

A question relevant to us here, is how a periodic function can be "represented" by a function series. What we mean by a representation in this case will hopefully be clear as we go on. The well known Taylor series will not be a good choice since a necessary condition here is that the function must be infinitely differentiable on its domain. The sawtooth function above clearly does not fulfill this when x takes integer values, $\dots - 2, -1, 0, 1, 2, \dots$. Even if the function has a Taylor series, this series is generally not a very practical choice since the Taylor series converges slowly for periodic functions. Then, we suggest to represent the periodic function $f : \mathbb{R} \supseteq S \rightarrow \mathbb{R}$ as a series of known periodic functions, that is,

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L} x\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L} x\right) . \quad (3.5)$$

The symbol " $\sim$ " is merely used here to symbolize a representation; we later define what we mean. This is the famous *Fourier series*, and the coefficients $a_0, a_1, a_2, \dots, b_1, b_2, b_3, \dots$ are called the *Fourier coefficients*, and n the *mode number*. If the Fourier series equals the function it represents we will apply the usual equal sign " $=$ ".

Example 12 The function $f(x) = 2 \cos(2\pi x/L) + 3 \sin(4\pi x/L)$ has two non-zero Fourier coefficients, namely, $a_2 = 2$ and $b_4 = 3$. $\diamond$

Example 13 The general solution in Equation (2.13) is a Fourier series with $a_n = 0$. $\diamond$

$n = 1$ is the *fundamental term* or *fundamental mode* and note that $\cos(\pi x/L)$ and $\sin(\pi x/L)$ have one period in the interval $[-L; L]$, that is, the period for the fundamental mode is $T = 2L$. For $n = 2$ the period is L , $n = 3$ the period is $2/3L$, and so forth; in general the relation between the mode period and L is

$$T = \frac{2L}{n} . \quad (3.6)$$

Often the Fourier series is expressed through the *wave vector* $k_n = 2\pi/T = n\pi/L$, that is,

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(k_n x) + \sum_{n=1}^{\infty} b_n \sin(k_n x) . \quad (3.7)$$

One may rightfully ask: "Exactly when can a periodic function be represented (whatever that means) by a Fourier series?". The more insightful answer to this question requires a rather in-depth discussion of series convergence which we will avoid. A sufficient set of conditions for us here is if

1. the function domain can be partitioned into a set of open intervals where the function is continuous, and
2. bounded on the domain, that is, there exists an $M > 0$ such that $|f(x)| \leq M$, for all $x \in S$.

The sawtooth function is clearly bounded. Moreover, we can partition each of the intervals $\dots, -1 \leq x < 0, 0 \leq x < 1, \dots$ into one open continuous interval $\dots, -1 < x < 0, 0 < x < 1, \dots$. Therefore, saw can be represented by a Fourier series. Importantly, the fact that the sawtooth function can be represented by a Fourier series does not mean that the two functions are equal in every point in the domain; we will later see that this is the case in the jump discontinuities.

To find the Fourier representation we must find the Fourier coefficients. To this end we need to know about *orthogonality* of functions. In general, let f and g be complex valued functions, $f, g : I \rightarrow \mathbb{C}$ where $I = [a; b] \subset \mathbb{R}$, then the *inner product* on I is defined as

$$\langle f, g \rangle = \int_a^b f(x) \bar{g}(x) dx, \quad (3.8)$$

where $\bar{g}$ is the complex conjugate to g . If f and g are real values functions we simply have $\langle f, g \rangle = \int_a^b f(x)g(x) dx$. We say that f and g are *orthogonal* on I if $\langle f, g \rangle = 0$.

Example 14 Let $f(x) = x$, $g(x) = \cos(x)$, and $h(x) = \sin(x)$. f and g are orthogonal on $I = [-\pi, \pi]$ because $\langle f, g \rangle = \int_{-\pi}^{\pi} x \cos(x) dx = 0$, but $f(x) = x$ and $h(x)$ are not orthogonal on I because $\langle f, h \rangle = \int_{-\pi}^{\pi} x \sin(x) dx = 2\pi \neq 0$. $\diamond$

We will use the following properties of the sine and cosine functions on $I = [-L; L]$

1. $\cos\left(\frac{n\pi x}{L}\right)$ and $\sin\left(\frac{m\pi x}{L}\right)$ are orthogonal for all $n, m \in \mathbb{N}$
2. $\cos\left(\frac{n\pi x}{L}\right)$ and $\cos\left(\frac{m\pi x}{L}\right)$ are orthogonal if $n \neq m$, otherwise if $n = m$ the inner product is L .
3. $\sin\left(\frac{n\pi x}{L}\right)$ and $\sin\left(\frac{m\pi x}{L}\right)$ are orthogonal if $n \neq m$, otherwise if $n = m$ the inner product is L .

These statements can be verified from direct calculations. For example,

$$\begin{aligned} \left\langle \cos\left(\frac{n\pi}{L}x\right), \sin\left(\frac{m\pi}{L}x\right) \right\rangle &= \int_{-L}^L \cos\left(\frac{n\pi}{L}x\right) \sin\left(\frac{m\pi}{L}x\right) dx \\ &= \frac{1}{2} \int_{-L}^L \sin\left(\frac{(n+m)\pi}{L}x\right) + \sin\left(\frac{(m-n)\pi}{L}x\right) dx \end{aligned} \quad (3.9)$$

using the product rule $2 \cos(a) \sin(b) = \sin(a+b) + \sin(a-b)$. This integral is always zero, hence, the functions are orthogonal.

Before this section's main theorem we need one additional definition

Definition Let $f : \mathbb{R} \supseteq S \rightarrow \mathbb{R}$ and let s_m be the partial sum

$$s_m(x) = \frac{a_0}{2} + \sum_{n=1}^m a_n \cos\left(\frac{n\pi}{L}x\right) + \sum_{n=1}^m b_n \sin\left(\frac{n\pi}{L}x\right). \quad (3.10)$$

We say that the Fourier series *converges uniformly* on S to f if for all $\epsilon > 0$ there exists an $N(\epsilon) \in \mathbb{N}$ such that $|s_m(x) - f(x)| < \epsilon$ for all $x \in S$ and $m \geq N$. $\circ$

Notice that N is independent of x . Uniform convergence entails that all points $x \in S$ converge with same upper bound N , hence, N only depends on ϵ not x . This is a stronger condition compared to point wise convergence where N in general is a function of x ; if a series converges uniformly to a function it also converges point wise.

We are now ready for the section's main theorem

Theorem If the Fourier series representation of $f : \mathbb{R} \rightarrow \mathbb{R}$ converges uniformly to f on $\mathbb{R}$ then

- 1 $\langle f(x), 1 \rangle = La_0$
- 2 $\langle f(x), \cos\left(\frac{n\pi}{L}x\right) \rangle = La_n$
- 3 $\langle f(x), \sin\left(\frac{n\pi}{L}x\right) \rangle = Lb_n$

on $I = [-L; L]$, and where $n \in \mathbb{N}_+$. These are the *Euler-Fourier equations*. $\circ$

Proof

1. Since the Fourier series converges uniformly to f the two function are equal, that is,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L}x\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L}x\right). \quad (3.11)$$

Integrating both sides from $-L$ to L

$$\int_{-L}^L f(x) dx = \frac{a_0}{2} \int_{-L}^L dx + \int_{-L}^L \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L}x\right) dx + \int_{-L}^L \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L}x\right) dx \quad (3.12)$$

Again due to the uniform convergence, we can interchange integration and summation in the last two terms. Then, for all $n \in \mathbb{N}_+$ we have

$$\int_{-L}^L \cos\left(\frac{n\pi}{L}x\right) dx = 0 \text{ and } \int_{-L}^L \sin\left(\frac{n\pi}{L}x\right) dx = 0, \quad (3.13)$$

and therefore

$$\int_{-L}^L f(x) dx = \frac{a_0}{2} \int_{-L}^L dx \Rightarrow La_0 = \int_{-L}^L f(x) dx. \quad (3.14)$$

This is the result we are after by definition of the inner product

$$\langle f(x), 1 \rangle = \int_{-L}^L f(x) dx \quad (3.15)$$

2. Multiplying Equation (3.11) with $\cos(m\pi x/L)$, $m = 1, 2, 3, \dots$ and integrate from $-L$ to L

$$\begin{aligned} \int_{-L}^L \cos\left(\frac{m\pi}{L}x\right) f(x) dx &= \frac{a_0}{2} \int_{-L}^L \cos\left(\frac{m\pi}{L}x\right) dx \\ &+ \int_{-L}^L \sum_{n=1}^{\infty} a_n \cos\left(\frac{m\pi}{L}x\right) \cos\left(\frac{n\pi}{L}x\right) dx \\ &+ \int_{-L}^L \sum_{n=1}^{\infty} b_n \cos\left(\frac{m\pi}{L}x\right) \sin\left(\frac{n\pi}{L}x\right) dx \end{aligned} \quad (3.16)$$

The first term on the right-hand side is zero. For the second term we again interchange the integration and summation, giving rise to integrals with relations

$$\int_{-L}^L \cos\left(\frac{m\pi}{L}x\right) \cos\left(\frac{n\pi}{L}x\right) dx = L\delta_{mn}, \quad (3.17)$$

which comes from the orthogonality properties above. δ_{mn} is the Kronecker delta, $\delta_{mn} = 1$ if $m = n$, otherwise if $m \neq n$ then $\delta_{mn} = 0$. This means that all terms are zero except for $m = n$. The third term yields integrals on the form

$$\int_{-L}^L \cos\left(\frac{m\pi}{L}x\right) \sin\left(\frac{n\pi}{L}x\right) dx = 0 \quad (3.18)$$

since cosine and sine are orthogonal on I .

Therefore, Equation (3.16) simplifies to

$$\int_{-L}^L \cos\left(\frac{m\pi}{L}x\right) f(x) dx = a_n \int_{-L}^L \cos\left(\frac{n\pi}{L}x\right) \cos\left(\frac{n\pi}{L}x\right) dx = La_n, \quad (3.19)$$

that is,

$$a_n = \frac{1}{L} \int_{-L}^L \cos\left(\frac{n\pi}{L}x\right) f(x) dx. \quad (n = 1, 2, 3, \dots) \quad (3.20)$$

Notice that if we allow $n = 0$ we arrive at Equation (3.14). This explains the choice of symbol for the constant term in the Fourier series.

The last equation is left as an exercise. •

The theorem for the Euler-Fourier coefficients requires uniform convergence which is a strong condition¹. One can, in fact, prove the theorem for less strict conditions; but we avoid this, and use the less satisfactory definition

Definition The Fourier representation of a function f is the Fourier series with coefficients given by the Euler-Fourier equations if they exist. ◦

Example 15 Find the Fourier coefficients of the sawtooth function. Then write-down the Fourier series representation.

The sawtooth function has jump discontinuities for every $x \in \mathbb{Z}$ and the Fourier series representation does not converge uniformly to the function. Nevertheless, we can find the Fourier representation.

a_0 : We use $L = 1$, then from Equation (3.14).

$$a_0 = \int_{-1}^1 \text{saw}(x) dx = \int_{-1}^0 (x+1) dx + \int_0^1 x dx = 1. \quad (3.21)$$

a_n : Rather than simply embarking on the tedious task of integrating Equation (3.20) we can re-arrange the integral, which leaves the problem straight forward. Substituting for saw we get

$$\begin{aligned} \int_{-1}^1 \text{saw}(x) \cos(n\pi x) dx &= \int_{-1}^0 (x+1) \cos(n\pi x) dx + \int_0^1 x \cos(n\pi x) dx \\ &= \int_{-1}^1 x \cos(n\pi x) dx + \int_{-1}^0 \cos(n\pi x) dx \end{aligned} \quad (3.22)$$

The first term is an integral over an odd function and this gives zero. For the last term we get

$$\int_{-1}^0 \cos(n\pi x) dx = \frac{1}{n\pi} \sin(n\pi x) \Big|_{-1}^0 = 0. \quad (3.23)$$

thus, $a_n = 0, n = 1, 2, 3, \dots$

b_n : Following the protocol above we have

$$\int_{-1}^1 \text{saw}(x) \sin(n\pi x) dx = \int_{-1}^1 x \sin(n\pi x) dx + \int_{-1}^0 \sin(n\pi x) dx \quad (3.24)$$

Using integration by parts the first integral is evaluated to

$$\int_{-1}^1 x \sin(n\pi x) dx = -\frac{1}{n\pi} x \cos(n\pi x) \Big|_{-1}^1 - \int_{-1}^1 \sin(n\pi x) dx \quad (3.25)$$

Sine is an odd function and the second integral is zero. Using the two relations

$$\cos(n\pi) = (-1)^n \text{ and } \cos(n\pi) = \cos(-n\pi) \quad (3.26)$$

¹and makes the theorem so-called *weak*

we then get $\int_{-1}^1 x \sin(n\pi x) = -2(-1)^n/n\pi$. The second integral in Equation (3.24) evaluates to

$$\int_{-1}^0 \sin(n\pi x) dx = -\frac{1 - (-1)^n}{n\pi} \quad (3.27)$$

and we get

$$b_n = -\frac{1 + (-1)^n}{n\pi}. \quad (3.28)$$

The Fourier series representation of the sawtooth function is therefore

$$\text{saw}(x) \sim \frac{1}{2} - \sum_{n=1}^{\infty} \frac{1 + (-1)^n}{n\pi} \sin(n\pi x) \quad (3.29)$$

Note that when n is odd the numerator in the summation is zero, and we can simply let n run over the even natural numbers. $\diamond$

From Equation (3.29) we see that for $x \in \mathbb{Z}$ the Fourier representation yields the value $1/2$, but this is not the value of the sawtooth function itself. It is a general feature of the Fourier representation that at point x_0 it has value of the average of the right and left limits of the function f it represents (so these must exist). That is, we have values

$$\frac{\lim_{x \rightarrow x_0^+} f(x) + \lim_{x \rightarrow x_0^-} f(x)}{2}, \quad x_0 \in \mathbb{R}. \quad (3.30)$$

In points where the function is continuous the Fourier representation value then equals the function value.

When visualizing the Fourier series we cannot plot the actual series as this means adding infinitely many terms. Rather we plot the partial sum

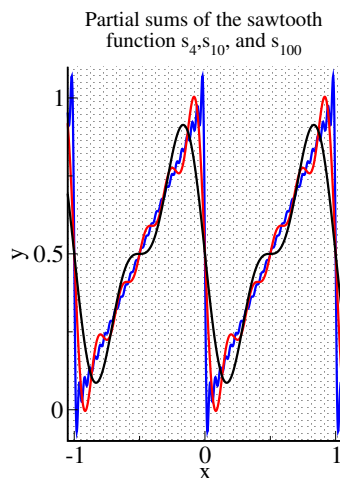
$$s_m(x) = \frac{1}{2} - \sum_{n=1}^m \frac{1 + (-1)^n}{n\pi} \sin(n\pi x). \quad (3.31)$$

We can define the remainder

$$R_m(x) = \text{saw}(x) - s_m(x). \quad (3.32)$$

The remainder R_{50} is plotted in Fig. 3.1 (a) for the sawtooth function. Notice a large disagreement in neighbourhoods around the points of discontinuities, which is manifested as large amplitude oscillations. This is known as the *Gibbs phenomenon*.

In Fig. 3.1 (b) the Fourier coefficients, b_n , are plotted as function of n (for n odd b_n is not shown). We see that the coefficients decrease, giving us the hope



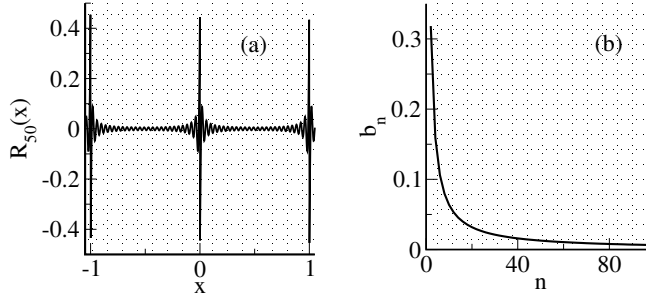


Figure 3.1: (a) The remainder function for the sawtooth function. (b) The Fourier coefficient, b_n , as function of n .

that the Fourier series actually converges. Needless to say, a convincing argument requires a proof, which we do not pursue here.

Function representations in terms of Fourier series can be viewed in terms of vector spaces. Consider the infinite dimensional vector space spanned by the set

$$\mathcal{V} = \{1, \cos(n\pi x/L), \sin(n\pi x/L) : n \in \mathbb{N}_+\} . \quad (3.33)$$

If we equip this vector space with the inner product, Eq. (3.8), $\mathcal{V}$ forms an orthogonal basis. The Fourier representation of a function is then the function represented in $\mathcal{V}$ by the Fourier coefficients. A vector space with inner product is also known as an inner product space having norm defined by the inner product $\|f\| = \sqrt{\langle f, f \rangle}$. More advanced treatments formalize this idea using Hilbert spaces.

Exercise 19

1. Prove that Equation (3.8) defines an inner product. (You may have to recall the exact definition of an inner product.)
2. Show that $f(x) = x$ and $g(x) = x^2 - 1$ are orthogonal on $I = [-1, 1]$, but not on $I = [-1; 2]$.
3. Prove that if $f \neq 0$ and $g \neq 0$ are orthogonal, then f and g are linearly independent. (Hint: You can use a contraction for this proof.)

Exercise 20 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and let the Fourier representation of f be uniform convergent to f . Prove the Euler-Fourier equation $\langle f(x), \sin(\frac{n\pi}{L}x) \rangle = Lb_n$

Exercise 21 Consider the function

$$f(x) = \begin{cases} 0 & -L \leq x < 0 \\ c_0 & 0 \leq x < L \end{cases} \quad (3.34)$$

1. Sketch a periodic expansion of f . Where do you expect the Gibbs phenomenon to arise?
2. Find the Fourier series representation of f .
3. Let $L = 1$. Plot the partial sums and the remainder for $m = 3, 9, 99$. (Why the odd index?)

Exercise 22 (Mandatory) An even function is defined as a function where $f(x) = f(-x)$ and an odd function a function where $f(x) = -f(-x)$. For example, cosine is even and sine odd. It is quite easy to show that, for example,

- the sum of even functions results in an even function,
- a sum of odd functions results in an odd function,
- if f is even

$$\int_{-L}^L f(x) dx = 2 \int_0^L f(x) dx$$

- if f is odd

$$\int_{-L}^L f(x) dx = 0$$

Therefore, if f is even the Fourier series is also even resulting in the *cosine series*

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L}x\right). \quad (3.35)$$

If f is odd the Fourier series is odd giving the *sine series*

$$\sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L}x\right). \quad (3.36)$$

Consider the function $f(x) = x^2$, $-2 < x < 2$. Find the Fourier series representation. Is the result consistent with the exercise preamble?

Exercise 23 Prove that the product of two odd functions is an even function. Is the product between two even functions then an odd function?

Exploration 2 In many "real life" situations you may not have a function definition for the periodic function or the Euler-Fourier integrals are untractable. Then the coefficients can be found by numerical methods. Go to Appendix 5.1.

3.2 The diffusion equation with Dirichlet boundaries

After Section 3.1 we are ready to explore the diffusion equation

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2}. \quad (3.37)$$

We will use the general IC

$$c(x, 0) = f(x), \quad (3.38)$$

and Dirichlet BCs

$$c(0, t) = c(L, t) = 0. \quad (3.39)$$

The starting point is that we assume c can be written as a product of two functions, X and T , which depends on x and t , respectively. That is,

$$c(x, t) = X(x)T(t). \quad (3.40)$$

This is referred to as *separation of variables*. Clearly, most function cannot be written in this form. The key observation is that sufficiently many solutions of this type exist to construct the general solution through superposition.

Question Can you give examples of functions that are not separable and some that are separable?

Inserting into the diffusion equation we obtain

$$\frac{\partial}{\partial t} X(x)T(t) = D \frac{\partial^2}{\partial x^2} X(x)T(t) \Rightarrow X(x) \frac{d}{dt} T(t) = DT(t) \frac{d^2}{dx^2} X(x). \quad (3.41)$$

If $T \neq 0$ for all $t > 0$ and $X \neq 0$ for interior points $0 < x < L$, we can rearrange this giving

$$\frac{1}{DT} \frac{dT}{dt} = \frac{1}{X} \frac{d^2 X}{dx^2}. \quad (3.42)$$

We have dropped the explicit dependencies on x and t .

Now, as t (the time) increases the function T will in general vary - how we do not know at the moment. However, the function X does not vary as t varies, so the right-hand side is a constant with respect to t . Likewise, as x varies the left-hand side is constant since T is independent of x . The constant must be the same, and we have

$$\frac{1}{DT} \frac{dT}{dt} = -\alpha \text{ and } \frac{1}{X} \frac{d^2 X}{dx^2} = -\alpha. \quad (3.43)$$

That we chose the constant to be $-\alpha$ is a matter of convenience. If $\alpha = 0$ we get the trivial time independent solution $X(x) = 0$ due to the boundary conditions which we avoid.

Before we continue, it is worth to emphasizing what just happened: Assuming that c can be factorized into a product of X and T we have transformed our (linear) partial differential equation into a problem of two (linear) ordinary differential equations for X and T ; and we can solve these two equations.

We start out by solving for X . Rearranging once again, we get

$$\frac{d^2 X}{dx^2} + \alpha X = 0. \quad (3.44)$$

We need the boundary conditions to continue. From Equation (3.39)

$$c(0, t) = X(0)T(t) = 0 \Rightarrow X(0) = 0, \quad (3.45)$$

when $T \neq 0$. The same is true for $x = L$, and we have Dirichlet BCs $X(0) = X(L) = 0$. We solved this exact boundary value problem in Section ??, where we could safely argue that $\alpha > 0$ since both the rate constant and the diffusion coefficient are positive. We cannot use this argument here, but since relation

$$\sqrt{\alpha} = \frac{n\pi}{L}, \quad (3.46)$$

is still valid we see that α must be real and positive. Then

$$\alpha = \left(\frac{n\pi}{L}\right)^2 \text{ with } n \in \mathbb{N}_+. \quad (3.47)$$

So for any $n \in \mathbb{N}_+$ we have the solution

$$X_n(x) = \chi_n \sin\left(\frac{n\pi}{L}x\right), \quad (3.48)$$

where χ_n is a constant (dependent on n).

α depends on n , and therefore the solution to T also depends on n . Thus, for any $n \in \mathbb{N}_+$ we have the problem

$$\frac{dT_n}{dt} = -\alpha D T_n, \quad (3.49)$$

with the general solution

$$T_n(t) = \tau_n e^{-(n\pi/L)^2 D t}. \quad (3.50)$$

We will not use the IC just yet.

From above we see that there are infinitely many solutions to c , namely,

$$c_n(x, t) = X_n(x)T_n(t) = b_n e^{-(n\pi/L)^2 D t} \sin\left(\frac{n\pi}{L}x\right), \quad (3.51)$$

where we have set $b_n = \chi_n \tau_n$. Invoking the superposition principle this means that the general solution is

$$c(x, t) = \sum_{n=1}^{\infty} c_n = \sum_{n=1}^{\infty} b_n e^{-(n\pi/L)^2 D t} \sin\left(\frac{n\pi}{L}x\right). \quad (3.52)$$

We still need a particular solution. We now use the IC, $c(x, 0) = f(x)$, that is, from the general solution given in Equation(3.52) we have at $t = 0$

$$c(x, 0) = f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L}x\right). \quad (3.53)$$

The IC is given by a Fourier sine series and the extension to $-L \leq x \leq L$ is an odd function. Thus, we can obtain the coefficients b_n from the Euler-Fourier equation

$$b_n = \frac{1}{L} \left\langle f(x), \sin\left(\frac{n\pi}{L}x\right) \right\rangle = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx \quad (3.54)$$

since the product of two odd functions is even, see exercise 22.

Question Can you show that Equation (3.52) is actually a solution to the diffusion equation with zero Dirichlet BCs?

Example 16 The solution to the diffusion equation with Dirichlet BCs and IC

$$c(x, 0) = c_0$$

is given by a sine series with Fourier coefficients

$$\begin{aligned} b_n &= \frac{2c_0}{L} \int_0^L \sin\left(\frac{\pi n}{L}x\right) dx \\ &= -\frac{2c_0}{n\pi}((-1)^n - 1). \end{aligned} \quad (3.55)$$

Notice that $b_n = 0$ for n even and $b_n = 4c_0/n\pi$ for n odd. Moreover, the IC is incompatible with the BCs, however, the solution still exists. $\diamond$

The result in Equation (3.52) is a central result and is used in the more general case where the Dirichlet BCs read

$$c(0, t) = c_1 \text{ and } c(L, t) = c_2. \quad (3.56)$$

First, we expect that for $t \rightarrow \infty$ the system reaches the steady state, which we denote c_{st} ; this follows by definition

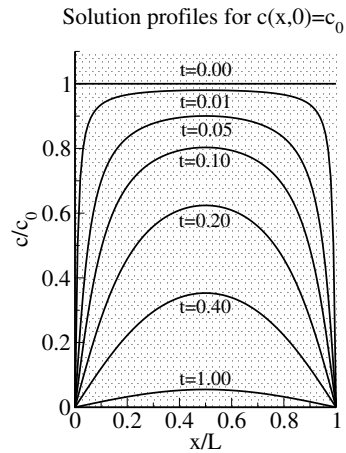
$$\frac{d^2 c_{st}}{dx^2} = 0, \quad (3.57)$$

with BCs in Equation (3.56). Using the method from Section 2 we can easily solve this giving

$$c_{st}(x) = ax + c_1 \quad (3.58)$$

where $a = (c_2 - c_1)/L$. We now define the function

$$\bar{c}(x, t) = c(x, t) - c_{st}(x), \quad (3.59)$$



where we subtract the steady-state case from c . The purpose of this transformation is to convert non-homogeneous boundary conditions into homogeneous boundary conditions. Note that

$$\frac{\partial c}{\partial t} = \frac{\partial \bar{c}}{\partial t} \quad (3.60)$$

and

$$\frac{\partial^2 c}{\partial x^2} = \frac{\partial^2 \bar{c}}{\partial x^2} - \frac{d^2 c_{\text{st}}}{dx^2} = \frac{\partial^2 \bar{c}}{\partial x^2}. \quad (3.61)$$

This means that $\bar{c}$ also fulfils the diffusion equation

$$\frac{\partial \bar{c}}{\partial t} = D \frac{\partial^2 \bar{c}}{\partial x^2}. \quad (3.62)$$

Moreover, the new function $\bar{c}$ has BCs

$$\bar{c}(0, t) = c(0, t) - c_{\text{st}}(0) = 0, \quad (3.63)$$

$$\bar{c}(L, t) = c(L, t) - c_{\text{st}}(L) = 0, \quad (3.64)$$

and IC

$$\bar{c}(x, 0) = c(x, 0) - c_{\text{st}}(x) = f(x). \quad (3.65)$$

This is just the problem we solved above! Therefore, the solution to $\bar{c}$ is just the Fourier sine series and, in turn, this means

$$c(x, t) = c_{\text{st}}(x) + \bar{c}(x, t) = ax + c_1 + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L}x\right) e^{-(n\pi/L)^2 Dt}, \quad (3.66)$$

where b_n is found directly from Equation (3.54)

$$b_n = \frac{2}{L} \int_0^L [c(x, 0) - c_{\text{st}}(x)] \sin\left(\frac{\pi n}{L}x\right) dx. \quad (3.67)$$

Exercise 24 Solve the diffusion equation on the domain $0 \leq x \leq 1$ with Dirichlet BCs

$$c(0, t) = c(1, t) = 0, \quad (3.68)$$

and IC $c(x, 0) = x$. In your favorite plotting program, plot the concentration profiles $c(x, t_0), c(x, t_1), \dots$ for a few different times.

Exercise 25 Solve the diffusion equation on the domain $0 \leq x \leq 1$ with Dirichlet BCs

$$c(0, t) = 0 \text{ and } c(1, t) = 1 \quad (3.69)$$

and IC $c(x, 0) = c_0$. Plot a few concentration profiles.

Exercise 26 If you have not done so already, show that Equation (3.52) actually is a solution to the diffusion equation with zero Dirichlet BCs. (Hint: Substitute the solution into the diffusion equation and compare the two sides of the equation.)

Exercise 27 (Mandatory) Consider the diffusion equation

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2}, \quad (3.70)$$

with IC $c(x, 0) = f(x)$ and Neumann BCs

$$\left. \frac{\partial c}{\partial x} \right|_{x=0} = \left. \frac{\partial c}{\partial x} \right|_{x=L} = 0. \quad (3.71)$$

Use separation of variable to show the solution to this is given by

$$c(x, t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n e^{-(n\pi/L)^2 Dt} \cos\left(\frac{n\pi}{L}x\right), \quad (3.72)$$

with coefficients

$$a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi}{L}x\right) dx. \quad (3.73)$$

for $n \in \mathbb{N}_+$.

Exploration 3 In the next chapter we deal with non-linear problems and sometimes we need to perform *supportive numerical explorations* of such systems. Since we now have the analytical solution for the (linear) diffusion equation, the time is right to implement a numerical scheme which can also handle general reaction-diffusion problems. Once we have checked the implementation against known solutions, we can, carefully, use the numerical method for non-linear problems as well. Go to Appendix 5.2.

3.3 Unbounded domains and Fourier transforms

Up until now we have dealt with Dirichlet, Neumann, or mixed boundary conditions. These boundaries characterise bounded systems (or domains), where we can clearly define some properties at the ends of the tube. In certain cases we have that the tube length is large compared to (i) the initial concentration profile and (ii) the characteristic *diffusion length scale*, $L_D(t) = \sqrt{4Dt}$. The second condition means that we limit our selves to study diffusion on sufficiently small times. We then apply *unbounded domain* BCs

$$\lim_{x \rightarrow \pm\infty} c(x, t) = 0. \quad (3.74)$$

Notice that for unbounded domains we shift the coordinate systems such that we have $-\infty < x < \infty$ rather than $0 \leq x < \infty$. As for bounded BCs we need some prerequisites before we can solve the diffusion equation with unbounded BCs.

First, we re-write the Fourier series in a complex form. Recall, in terms of wave vector k_n the Fourier series is given by

$$F(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(k_n x) + \sum_{n=1}^{\infty} b_n \sin(k_n x). \quad (3.75)$$

We have the following relations

$$\begin{aligned} \cos(k_n x) &= \frac{e^{ik_n x} + e^{-ik_n x}}{2} \\ \sin(k_n x) &= \frac{e^{ik_n x} - e^{-ik_n x}}{2i} \end{aligned} \quad (3.76)$$

which can be shown directly by Euler's identity. Substitution leads to

$$\begin{aligned} F(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} \frac{a_n}{2} (e^{ik_n x} + e^{-ik_n x}) + \sum_{n=1}^{\infty} \frac{b_n}{2i} (e^{ik_n x} - e^{-ik_n x}) \\ &= \frac{a_0}{2} + \sum_{n=1}^{\infty} \frac{ia_n + b_n}{2i} e^{ik_n x} + \sum_{n=1}^{\infty} \frac{ia_n - b_n}{2i} e^{-ik_n x}. \end{aligned} \quad (3.77)$$

Note, we here assumed that we can interchange the terms in the series. The wave vector has the property that

$$k_{-n} = \frac{-n\pi}{L} = -k_n \quad (3.78)$$

and therefore we can change the index of the last term in Equation (3.77)

$$\sum_{n=1}^{\infty} \frac{ia_n - b_n}{2i} e^{-ik_n x} = \sum_{n=-\infty}^{-1} c_n e^{ik_n x}. \quad (3.79)$$

Inserting into Equation (3.77)

$$F(x) = \sum_{n=-\infty}^{\infty} c_n e^{ik_n x} \quad (3.80)$$

where $c_n \in \mathbb{C}$ are the *complex Fourier coefficients* which are related to the real Fourier coefficients by

$$c_n = \begin{cases} \frac{ia_{|n|} - b_{|n|}}{2i} & \text{if } n < 0 \\ \frac{ia_n + b_n}{2i} & \text{if } n > 0 \\ \frac{a_0}{2} & \text{if } n = 0 \end{cases} \quad (3.81)$$

The complex Fourier coefficients are found from the orthogonal property of the exponential function on the usual interval $[-L; L]$.

Theorem If the complex Fourier series representation of $f : \mathbb{R} \rightarrow \mathbb{R}$ converges uniformly to f on $\mathbb{R}$ and $f = F$ then

$$\langle f(x), e^{-ik_n x} \rangle = 2Lc_n \quad (3.82)$$

on $I = [-L; L]$. Moreover, $\langle f(x), e^{-ik_m x} \rangle = 0$ on I if $m \neq n$. $\circ$

Proof See exercise $\bullet$

In unbounded domains we cannot general in resolve the function in periods of finite L . The (complex) Fourier coefficients are then defined to be the limiting value of

$$c_n = \lim_{L \rightarrow \infty} \frac{1}{2L} \int_{-L}^L f(x) e^{-ik_n x} dx \quad (3.83)$$

if the integral exists. This definition is not as innocent as it may first appear since the wave vector is dependent on both the index n and L , and the limit is not always easy to evaluate even if the integral can be found. However, Equation (3.83) motivates another very powerful definition, namely, the definition of the *Fourier transform*

$$\tilde{f}(k) = \int_{-\infty}^{\infty} f(x) e^{-ikx} dx. \quad (3.84)$$

We here consider k to be any positive real number and we omit the subscript n . The Fourier transform is an *integral transform* or *integral operator* that transforms the function f into another function $\tilde{f}$ that depends on the wave vector k .

Example 17 Find the Fourier transform $\tilde{f}$ for

$$f(x) = \begin{cases} e^{-ax} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases} \quad (3.85)$$

where $a > 0$. From the definition

$$\begin{aligned} \tilde{f}(k) &= \int_{-\infty}^{\infty} e^{-ax} e^{-ikx} dx = \int_0^{\infty} e^{-(a+ik)x} dx \\ &= -\frac{1}{a+ik} [e^{-ax} (\cos(kx) - i \sin(kx))]_0^{\infty} = \frac{1}{a+ik}. \end{aligned} \quad (3.86)$$

Notice $\tilde{f}$ is a complex valued function. $\diamond$

The Fourier transform is often written in terms of the operator symbol $\mathcal{F}$, that is,

$$\tilde{f}(k) = \mathcal{F}[f](k). \quad (3.87)$$

The Fourier transform has some very powerful properties which we will exploit.

Theorem

1 If $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are bounded and piecewise continuous, then

$$\mathcal{F}[\alpha f + g] = \alpha \mathcal{F}[f] + \mathcal{F}[g] \quad (3.88)$$

for $\alpha \in \mathbb{R}$. This is the linearity property.

2 If $f : \mathbb{R} \rightarrow \mathbb{R}$ is at least twice differentiable, bounded and decays sufficiently quickly to zero as $x \rightarrow \infty$, then

$$\mathcal{F}\left[\frac{df}{dx}\right] = ik\mathcal{F}[f] \quad \text{and} \quad \mathcal{F}\left[\frac{d^2f}{dx^2}\right] = -k^2\mathcal{F}[f] \quad (3.89)$$

3 Let $f = f(x, t)$ be defined for all positive t and $x \in \mathbb{R}$. Assume that $f \in C^1$ and the Leibniz integral rule applies, then

$$\mathcal{F}\left[\frac{\partial f}{\partial t}\right] = \frac{\partial}{\partial t}\mathcal{F}[f]. \quad (3.90)$$

◦

Proof

1: The linearity follows directly from the linearity for integrals.

2: The second property can be shown by integration by parts

$$\begin{aligned} \mathcal{F}\left[\frac{df}{dx}\right] &= \int_{-\infty}^{\infty} f'(x)e^{-ikx} dx \\ &= [f(x)e^{-ikx}]_{-\infty}^{\infty} + ik \int_{-\infty}^{\infty} f(x)e^{ikx} dx = ik\tilde{f}(k) \end{aligned} \quad (3.91)$$

since $\lim_{x \rightarrow \pm\infty} f = 0$. For the second order derivative integration by parts is applied twice.

3: For the third property we define the function $w(x, t) = f(x, t)e^{-ikx}$. Since the Leibniz integral rule exists we get

$$\frac{\partial}{\partial t}\mathcal{F}[f] = \frac{\partial}{\partial t} \int_{-\infty}^{\infty} w(x, t) dx = \int_{-\infty}^{\infty} \frac{\partial w}{\partial t} dx = \int_{-\infty}^{\infty} \frac{\partial f}{\partial t} e^{-ikx} dx \quad (3.92)$$

which is the left-hand side of Equation (3.90). •

We can use the properties of the Fourier transform to the diffusion equation with unbounded BCs. Letting the operator act on both sides

$$\mathcal{F}\left[\frac{\partial c}{\partial t}\right] = \mathcal{F}\left[D\frac{\partial^2 c}{\partial x^2}\right] \quad (3.93)$$

implies that

$$\frac{\partial}{\partial t} \tilde{c}(k, t) = -Dk^2 \tilde{c}, \quad (3.94)$$

for all k . In this way, we transform the partial differential equation for c into a differential equation for $\tilde{c}$, but with respect to a single variable. The general solution we know

$$\tilde{c}(k, t) = \tilde{c}_0 e^{-Dk^2 t}. \quad (3.95)$$

This is referred to as *the solution in Fourier space*².

If we wish a particular solution we of course need to specify the initial condition. For example, if we have an initial concentration which is Gaussian distributed with respect to x

$$c(x, 0) = c_0 e^{-(x/\sigma)^2}, \quad (3.96)$$

where $\sigma > 0$ is a measure of the Gaussian width and c_0 the concentration at $x = 0$, then

$$\tilde{c}_0 = c_0 \int_{-\infty}^{\infty} e^{-(x/\sigma)^2 - ikx} dx. \quad (3.97)$$

The trick is now to rewrite the exponent into another form

$$-\frac{x^2}{\sigma^2} - ikx = -\frac{1}{\sigma^2} (x^2 + ik\sigma^2 x) = -\frac{1}{\sigma^2} \left[\left(x + \frac{ik\sigma^2}{2} \right)^2 + \frac{k^2\sigma^4}{4} \right]. \quad (3.98)$$

Substitution and the Fourier transform now reads

$$\tilde{c}_0 = c_0 e^{-\frac{k^2\sigma^2}{4}} \int_{-\infty}^{\infty} e^{-\frac{1}{\sigma^2} \left(x + \frac{ik\sigma^2}{2} \right)^2} dx = c_0 \sqrt{\pi} \sigma e^{\frac{k^2\sigma^2}{4}} \quad (3.99)$$

applying the useful Gaussian integral formula

$$\int_{-\infty}^{\infty} e^{-a(x+b)^2} dx = \sqrt{\frac{\pi}{a}}. \quad (3.100)$$

Substitution into Equation (3.95) we get

$$\tilde{c}(k, t) = c_0 \sqrt{\pi} \sigma e^{\frac{k^2\sigma^2}{4}} e^{-Dk^2 t} \quad (3.101)$$

Sometimes the solution in Fourier space can give us the information we seek, however, we often also want the solution to c and not only $\tilde{c}$. We now define the *inverse Fourier transform* of $\tilde{f}$ to be

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx} dk. \quad (3.102)$$

²We have not formally defined this "space", but we will use the term nevertheless.

In operator notation we write this as $f(x) = \mathcal{F}^{-1}[\tilde{f}](x)$. Let us see a useful example.

Example 18 Let $f(x) = e^{-ax^2}$, where $a > 0$. Rewriting the exponent and using Gaussian's integration formula

$$\tilde{f}(k) = \sqrt{\frac{\pi}{a}} e^{-k^2/4a}. \quad (3.103)$$

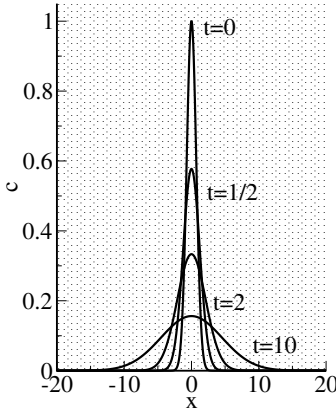
The inverse Fourier transform of $\tilde{f}$ is

$$\begin{aligned} \frac{1}{2\sqrt{a\pi}} \int_{-\infty}^{\infty} e^{-k^2/4a+ikx} dk &= \frac{e^{-ax^2}}{2\sqrt{a\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{4a}(k+2aix)^2} dk \\ &= \frac{\sqrt{4a\pi}}{2\sqrt{a\pi}} e^{-ax^2} = e^{-ax^2} \end{aligned} \quad (3.104)$$

using the identity

$$-k^2/4a + ikx = -\frac{1}{4a} (k^2 - i4akx) = -\frac{1}{4a} (k + i2ax)^2 - ax^2. \quad (3.105)$$

Notice that we recover the original function $f(x) = e^{-ax^2}$. $\diamond$



The example is not a coincidence, of course, and is generalized in the following statement: If $f : \mathbb{R} \rightarrow \mathbb{R}$ is sufficiently well behaved, then

$$\mathcal{F}^{-1}[\mathcal{F}[f]](x) = \mathcal{F}^{-1}[\tilde{f}](x) = f(x). \quad (3.106)$$

This is the important *inverse Fourier theorem*.

Equation (3.101) is continuous and bounded thus the inverse Fourier theorem guarantees that we recover c by an inverse transform. We can rewrite Equation (3.101) to

$$\tilde{c}(k, t) = c_0 \sqrt{\pi} \sigma e^{-\alpha k^2}, \quad (3.107)$$

with $\alpha = \frac{\sigma^2}{4} + Dt$. This is of the same form as Equation (3.103), and we can simply substitute and get

$$c(x, t) = \frac{c_0 \sigma}{\sqrt{\sigma^2 + 4Dt}} e^{-x^2/(\sigma^2 + 4Dt)}. \quad (3.108)$$

This is the solution³ to the diffusion equation in an unbounded domain with initial conditions given by Equation (3.96).

Exercise 28 Confirm Equation (3.108).

³finally..!

Exercise 29 Prove the theorem on page 52.

Exercise 30 Find the Fourier transform for

$$f(x) = \begin{cases} -x^2 + 1 & -1 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad (3.109)$$

and plot the function.

Exploration 4 Recall the diffusion equation in polar coordinates and with angular symmetry

$$\frac{\partial c}{\partial t} = \frac{D}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial c}{\partial \rho} \right). \quad (3.110)$$

Envision an experiment where we add a drop of a chemical substance into the center of a Petri dish that is filled with a solvent. We assume that the radius of the Petri dish is large compared to initial substance concentration distribution and that the diffusion length scale L_D is small.

When can we use Equation (3.108) as an approximation to the solution to this diffusion system? Solve Equation (3.110) numerically using the FCTS algorithm and compare with your answer.

Chapter 4

Reaction-diffusion phenomena

We have, separately, treated the steady-state case and the non-steady pure diffusive case. This will help us in this section, where we put it together. We start with the tractable linear case and then move on to some examples of non-linear systems and phenomena.

4.1 Reaction-diffusion in pores and channels: The linear case

Recall, the simple linear growth rate system presented in Example 1

$$\frac{\partial c}{\partial t} = kc + D \frac{\partial^2 c}{\partial x^2}, \quad (4.1)$$

with IC and BCs

$$c(x, 0) = f(x) \text{ and } c(0, t) = c(L, t) = 0. \quad (4.2)$$

The trick here is to transform the function we seek, c , into another function, say y , via an exponential transformation

$$y(x, t) = c(x, t)e^{-kt}. \quad (4.3)$$

The relations between the derivatives of y and c are readily found to be

$$\frac{\partial c}{\partial t} = ke^{kt}y + e^{kt}\frac{\partial y}{\partial t} \text{ and } \frac{\partial^2 c}{\partial x^2} = e^{kt}\frac{\partial^2 y}{\partial x^2}. \quad (4.4)$$

Then, by substitution and cancelling the exponential factors, we recover a simple diffusion equation, but now with respect to y

$$\frac{\partial y}{\partial t} = D \frac{\partial^2 y}{\partial x^2}. \quad (4.5)$$

From the transformation we see that the IC for y reads

$$y(x, 0) = c(x, 0) = f(x) \quad (4.6)$$

The left BC is

$$y(0, t) = c(0, t)e^{-kt} \implies y(0, t) = 0 \text{ since } c(0, t) = 0. \quad (4.7)$$

Likewise the right BC is

$$y(L, t) = c(L, t)e^{-kt} \implies y(L, t) = 0 \text{ since } c(L, t) = 0. \quad (4.8)$$

We know the solution to this problem from the previous chapter

$$y(x, t) = \sum_{n=1}^{\infty} b_n e^{-D(n\pi/L)^2 t} \sin\left(\frac{n\pi x}{L}\right), \quad (4.9)$$

where the Fourier coefficient b_n is given by the Euler-Fourier equation

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx. \quad (4.10)$$

To recover the function c we simply use the inverse transformation $c(x, t) = y(x, t)e^{kt}$, and after substitution we get

$$c(x, t) = \sum_{n=1}^{\infty} b_n e^{(k-D(n\pi/L)^2)t} \sin\left(\frac{n\pi x}{L}\right). \quad (4.11)$$

Like the diffusion equation with Dirichlet BCs, the solution to this linear RD-equation is a sine series. However, each term in the series now has a temporal dynamics that depends on both the reaction rate coefficient k and the diffusion coefficient D . From the exponent we immediately conclude that each term (or Fourier mode)

$$\begin{aligned} &\text{decays exponentially if } k < D(n\pi/L)^2 \\ &\text{is constant if } k = D(n\pi/L)^2 \\ &\text{increases exponentially if } k > D(n\pi/L)^2 \end{aligned} \quad (4.12)$$

In particular, all modes decay if $k < 0$.

Example 19 Let $f(x) = \sin(\pi x/L)$, then

$$b_n = \frac{2}{L} \int_0^L \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx = \begin{cases} 1 & \text{if } n = 1 \\ 0 & \text{otherwise} \end{cases} \quad (4.13)$$

and therefore only the mode $n = 1$ is non-zero, i.e.,

$$c(x, t) = e^{(k-D(\pi/L)^2)t} \sin\left(\frac{\pi x}{L}\right). \quad (4.14)$$

If $k < D(\pi/L)^2$ the initial sinusoidal profile decays to the trivial steady-state solution $c = 0$. If $k = D(\pi/L)^2$ the initial profile is the steady-state profile, whereas $k > D(\pi/L)^2$ yields an exponential growth at any point x in the domain from 0 to L . $\diamond$

Question How do the relations in (4.12) pertain to the discussion on page 15?

Exercise 31 The rate constant k has unit of per time, and the diffusion coefficients has units of length squared per time. Use these units to transform the linear RD-equation into the dimensionless equation. You can assume c represents a population size and is dimensionless.

Exercise 32 Show the solution to Equation (4.1) with Neumann BCs

$$\left. \frac{\partial c}{\partial x} \right|_{x=0} = \left. \frac{\partial c}{\partial x} \right|_{x=L} = 0 \quad (4.15)$$

always diverges if $k > 0$ and $c(x, 0) > 0$. Give a bio-chemical interpretation of this dynamics. Hint: You can use the result from Exercise 27.

Exercise 33 (Prerequisite: Exploration 3) Perform a numerical exploration to confirm the relations in Equation (4.12) using IC $f(x) = \sin(\pi x/L)$.

4.2 Propagating fronts

Consider two molecular species, say, A and B. These two types of molecules can react giving two A molecules. We write the reaction mechanism as



Since A catalyses its own production this is an example of an *autocatalytic reaction*. A real chemical reaction featuring autocatalytic behavior is the iodate-iodine reaction, see Fig. 4.1.

The reaction rate can be modelled by applying *mass action kinetics* such that

$$r = k' c_A c_B, \quad (4.17)$$

where $k' > 0$ is the (second order) reaction rate coefficient, and c_A and c_B are the concentration of A and B, respectively. The rate of change for A is then

$$\frac{dc_A}{dt} = k' c_A c_B. \quad (4.18)$$

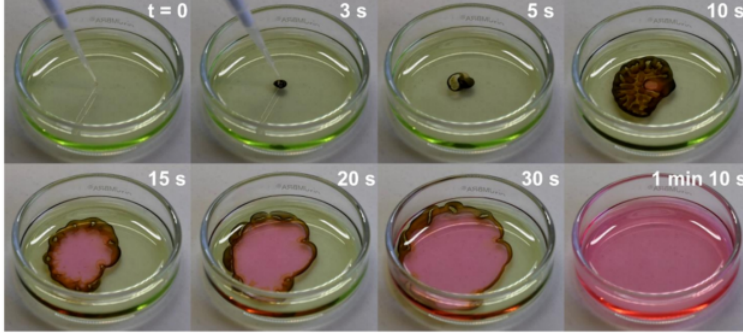


Figure 4.1: A drop of an iodine solution is placed in an iodate solution. The reaction is autocatalytic with respect to iodine, and a clear chemical front is observed where the iodate solution is ahead of the front and the rich iodine solution behind the front. Picture is from Panzarasa, *Reaction Kinetics, Mechanisms and Catalysis*, 135:1349 (2022).

Let $c_0 = c_A + c_B$ be the total concentration and assume this is a constant, then

$$\frac{dc_A}{dt} = k'c_A(c_0 - c_A). \quad (4.19)$$

Allowing for diffusion and dividing by c_0 we have

$$\frac{\partial c}{\partial t} = kc(1 - c) + D \frac{\partial^2 c}{\partial x^2}, \quad (4.20)$$

where $c = c_A/c_0$ and $k = k'c_0$. This is the *Fisher-Kolmogorov equation* or *logistic growth with diffusion*. Notice that c is bounded, $0 \leq c \leq 1$.

Question Exactly, how did we arrive at Equation (4.18) from Equation (4.17)?

Envision that we have a one dimensional system (i.e., a thin tube) composed of only reactive B molecules (and likely also an inert solvent). At time $t = 0$ we introduce some A molecules at the left boundary, $x = 0$. Encounters between A and B molecules result in the autocatalytic production of A, leading to a local concentration gradient and, hence, a flux where A molecules diffuse into rich B regions (positive x -direction), where, again, A molecules are produced, and so forth. This, we may imagine, leads to a propagating concentration front of A moving through the system in the positive x -direction. Behind the front we have A molecules, and ahead of the front B molecules; in the front zone both A and B molecules are present which react according to (4.16). This is illustrated in Fig. 4.2(a).

One research question we now seek to answer is: *If* the propagating front exists and *if* the speed is constant, how fast will the front propagate?

Assume that (after a transient time) the front propagates with a constant speed, s . We can then choose to shift the coordinate system with the front speed. In this

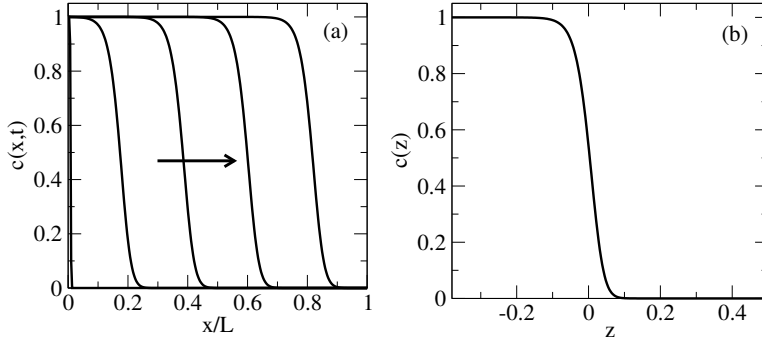


Figure 4.2: Propagating front in the standard (or fixed) coordinate system, (a), and in the moving coordinate frame, (b). The arrow in (a) indicates the direction of the front propagation (positive x -direction).

moving frame the concentration profile is independent of time. (Imagine you 'sit' on the front moving along with it, then you observe it being independent of time. This is not the case for an observer standing in a fixed coordinate system, where the front will pass her at some point in time.) Since the front propagates in the positive x -direction we define the moving frame coordinate by

$$z = x - st. \quad (4.21)$$

The concentration profile can now be written as depending on z ; $c(z) = c(z(x, t)) = c(x, t)$. See Fig. 4.2 (b). Importantly, we must expect

$$\lim_{z \rightarrow -\infty} c = 1 \text{ and } \lim_{z \rightarrow \infty} c = 0. \quad (4.22)$$

This means that in the moving frame $c = 1$ acts as an unstable fixed point and that $c = 0$ as a stable fixed point. In our treatment here, we will require that for a front to exist there must be a z_0 such that

$$\left. \frac{dc}{dz} \right|_{z=z_0} \neq 0 \text{ and } \lim_{z \rightarrow \pm\infty} \frac{dc}{dz} = 0. \quad (4.23)$$

To proceed we need to derive the dynamical equation for the concentration in the moving coordinate system. From the chain rule

$$\frac{\partial c}{\partial t} = -s \frac{dc}{dz} \text{ and } \frac{\partial^2 c}{\partial x^2} = \frac{d^2 c}{dz^2}. \quad (4.24)$$

Substitution into the Fisher-Kolmogorov equation gives

$$D \frac{d^2 c}{dz^2} + s \frac{dc}{dz} + kc(1 - c) = 0. \quad (4.25)$$

Remember that $k > 0$ since $k' > 0$.

This is an important result: As with the diffusion equation, we have managed to transform a partial differential equation into a problem of an ordinary differential equation. This is extremely helpful because we can now analyse this using standard techniques. Also notice that s , which we hope to say something about, enters the problem.

In our analysis of the dynamics we start with a simple linear stability analysis. First, we recast the second order differential equation into a system of first-order equations

$$\begin{aligned}\frac{dc}{dz} &= v \\ D\frac{dv}{dz} &= -sv - kc(1-c).\end{aligned}\tag{4.26}$$

where v is an auxillary variable. The fixed points are

$$(c, v) = (0, 0) \text{ and } (c, v) = (1, 0).\tag{4.27}$$

From the previous discussion we expect that $(0, 0)$ is stable and $(1, 0)$ is unstable, but sometimes our expectations will betray us, and we will now show this. The general Jacobian matrix is (please, check!)

$$\mathbf{J} = \begin{bmatrix} 0 & 1 \\ k(2c-1)/D & -s/D \end{bmatrix}.\tag{4.28}$$

For the fixed point $(1, 0)$ we have the eigenvalues

$$\lambda_{1,2} = -\frac{1}{2} \left(\frac{s}{D} \pm \sqrt{\left(\frac{s}{D}\right)^2 + \frac{4k}{D}} \right).\tag{4.29}$$

Since $k > 0$ and $D > 0$, the sum under the square root is positive and the square root term is larger than s/D . This means that the eigenvalues are real, one being negative and one positive. The point $(1, 0)$ is then an unstable saddle point.

For the fixed point $(0, 0)$ we have the eigenvalues

$$\lambda_{1,2} = -\frac{1}{2} \left(\frac{s}{D} \pm \sqrt{\left(\frac{s}{D}\right)^2 - \frac{4k}{D}} \right).\tag{4.30}$$

The three possible scenarios are

1. If $(s/D)^2 > 4k/D$, then the sum under the square root is positive and $\left[\left(\frac{s}{D}\right)^2 - \frac{4k}{D}\right]^{1/2} < s/D$, that is, the eigenvalues are both real and negative; the fixed point is a stable node.
2. If $(s/D)^2 = 4k/D$ we have repeated eigenvalues $\lambda_{1,2} = -s/2D$ and the fixed point is a stable node.
3. If $(s/D)^2 < 4k/D$ the fixed point has two complex conjugate eigenvalues with negative real parts, that is, the fixed point is a stable spiral.

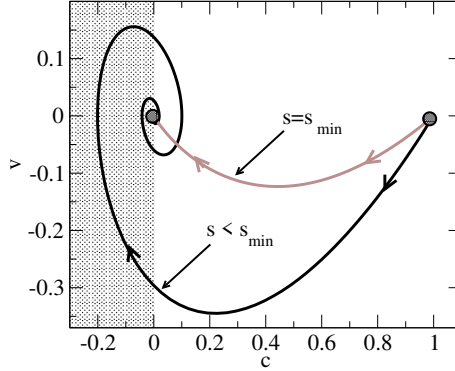


Figure 4.3: Two phase plane trajectories starting close to the unstable saddle point $(1,0)$. The two examples are when $s = s_{\min}$ and when $s < s_{\min}$.

The last scenario is problematic when we interpret the function c as being a normalized concentration since we then require $c \geq 0$. Recall, $c(z) = c(x, t) = c_A/c_0 \geq 0$. If $(0,0)$ is a stable spiral the phase plane trajectories will spiral around the origin $(0,0)$ and c will take negative values. Therefore, the eigenvalues cannot be complex, implying that only scenario 1 and 2 are useful. This in turn defines a minimum allowed speed

$$\left(\frac{s}{D}\right)^2 \geq \frac{4k}{D} \Rightarrow s \geq 2\sqrt{kD} = s_{\min}. \quad (4.31)$$

In Fig. (4.3) the trajectories are shown for $s < s_{\min}$ (the stable spiral case) and $s = s_{\min}$ (the stable node case).

Important From the linear stability analysis we have only derived a minimum allowed front propagation speed, not the actual speed the front takes. This is, furthermore, under the assumption that the front exists and propagates with constant speed. To investigate the system further we will rely on numerical analysis, see Exploration 5.

We can do a bit more. Again, we will assume that the front exists and that it propagates with constant speed. Now, Equation (4.25) is a non-linear problem, which we cannot solve in general. However, we can find an approximate solution using a perturbation method. To this end, we identify a small perturbation parameter $\epsilon = 1/s^2$, that is, we focus our analysis on the limit of large front speeds. The choice of perturbation parameter is clear when we perform a second coordinate transformation. Let

$$\xi = kz\sqrt{\epsilon}. \quad (4.32)$$

From the chain rule we have the following identities

$$\frac{dc}{dz} = k\sqrt{\epsilon} \frac{dc}{d\xi} \quad \text{and} \quad \frac{d^2c}{dz^2} = k^2\epsilon \frac{d^2c}{d\xi^2}. \quad (4.33)$$

Substituting into Equation (4.25), we get

$$\epsilon k D \frac{d^2 c}{d\xi^2} + \frac{dc}{d\xi} + c(1 - c) = 0. \quad (4.34)$$

This is a singular perturbation problem, since the case $\epsilon = 0$ yields a first order differential equation, which in general is qualitatively different from the original differential equation. We boldly attempt to continue using the (naive) outer solution which we introduced for regular perturbation problems.

We substitute the Poincaré expansion

$$c(\xi) \sim \sum_{n=0}^N \epsilon^n c_n(\xi) \quad (4.35)$$

into Equation (4.34) and collect terms of equal order in ϵ . To zeroth order we obtain

$$\frac{dc_0}{d\xi} + c_0(1 - c_0) = 0. \quad (4.36)$$

This is a first order equation and, also, note it does not contain the diffusion coefficient. This implies that we look at the effect of the reaction only, but under the assumption that the front actually exists, hence, the diffusion process is implicitly included. Separation of variables gives

$$\frac{dc_0}{c_0(1 - c_0)} = -d\xi. \quad (4.37)$$

The fraction on the left-hand side can be written as

$$\frac{1}{c_0(1 - c_0)} = \frac{1}{c_0} + \frac{1}{1 - c_0}, \quad (4.38)$$

and integration gives

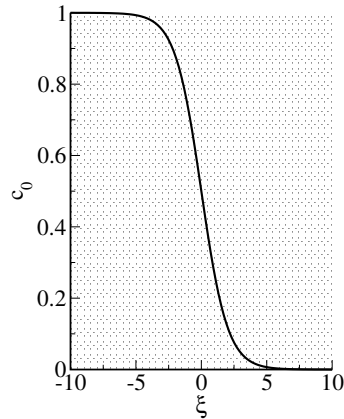
$$\ln \left| \frac{c_0}{1 - c_0} \right| = -\xi + A, \quad (4.39)$$

where A is an integration constant. Rearranging

$$\frac{c_0}{1 - c_0} = e^A e^\xi \implies c_0 = \frac{1}{1 + K e^\xi}. \quad (4.40)$$

Here we have $K = e^{-A}$. We choose the IC $c_0(0) = 1/2$ giving $K = 1$ and therefore

$$c_0 = \frac{1}{1 + e^\xi} = \frac{1}{1 + e^{kz/s}}. \quad (4.41)$$



This choice of IC is not unique; different choices will just shift the concentration profile in ξ -direction. If $s = s_{\min}$ we get

$$c_0 = \frac{1}{1 + e^{\sqrt{k/(4D)}z}}. \quad (4.42)$$

We continue the analysis in the exercises.

The idea of propagating fronts is used to model different systems. Let us see a two-species predator-prey example.

Example 20 This example is from Murray, *Mathematical Biology*; I will leave out quite a few details and recommend that you follow along with pen and paper.

We consider predator movement from regions of high population of predators into low populated regions. The latter region is rich in prey due to the low predator concentration. This version of the model assumes that the prey does not perform any migration. Let $0 \leq c_1 \leq 1$ be the dimensionless prey concentration and $0 \leq c_2 \leq 1$ the dimensionless predator concentration. The model in dimensionless form reads

$$\frac{\partial c_1}{\partial t} = c_1(1 - c_1 - c_2), \quad \frac{\partial c_2}{\partial t} = ac_2(c_1 - b) + \frac{\partial^2 c_2}{\partial x^2}, \quad (4.43)$$

where the model parameters satisfy $a, b > 0$.

We assume that there exists a front and that the front speed is constant. We may once more ask the question “If the front exists, what is the minimum allowed propagation speed?”. We approach the problem exactly as we did for the Fisher-Kolmogorov front, but, just for fun, in this example we let the front propagate from right to left (i.e., negative x -direction). This means that the moving frame coordinate is given by

$$z = x + st. \quad (4.44)$$

Let $c_1(z) = c_1(x, t)$ and $c_2(z) = c_2(x, t)$. Then in the co-moving frame we have the dynamics

$$s \frac{dc_1}{dz} = c_1(1 - c_1 - c_2) \quad (4.45)$$

$$s \frac{dc_2}{dz} = ac_2(c_1 - b) + \frac{d^2 c_2}{dz^2} \quad (4.46)$$

from which we can form a system of first order differential equations

$$\begin{aligned} \frac{dc_1}{dz} &= \frac{c_1}{s}(1 - c_1 - c_2) \\ \frac{dc_2}{dz} &= w \\ \frac{dw}{dz} &= sw - ac_2(c_1 - b) \end{aligned} \quad (4.47)$$

Again, w simply acts as an auxiliary variables. The fixed points for this are

$$\begin{aligned} \text{unstable point :} & \quad (c_1, c_2, w) = (0, 0, 0) \\ \text{unstable point :} & \quad (c_1, c_2, w) = (1, 0, 0) \\ \text{stable point :} & \quad (c_1, c_2, w) = (b, 1 - b, 0) \end{aligned}$$

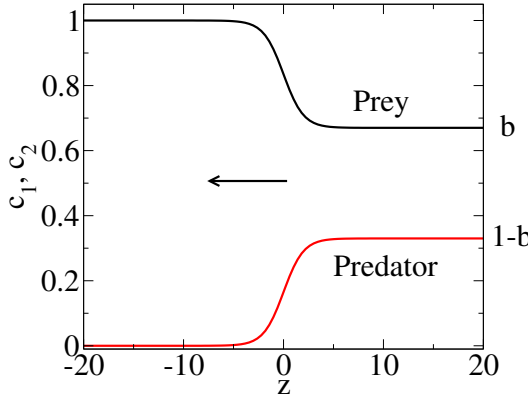


Figure 4.4: Illustration of the propagating front system in Example 20. Front propagation in the fixed frame is indicated by the arrow. $b = 2/3$

Since $0 \leq c_1, c_2 \leq 1$ we get the constraint that $0 < b < 1$. The first fixed point is simply the trivial case, and is not interesting here. The two other points are more informative: As $z \rightarrow -\infty$ we approach the unstable fixed point $(1, 0, 0)$, that is, there is only prey ahead of the front. As $z \rightarrow \infty$ we approach the stable fixed point $(b, 1 - b, 0)$, this is the co-existence state between the prey and predator, having concentrations b and $1 - b$, respectively. See illustration in Fig. 4.4.

We demand that the concentrations are positive and bounded so $0 \leq c_1, c_2 \leq 1$. We can then attempt the same analysis as we did for the Fisher-Kolmogorov system, but this time for the unstable fixed point $(1, 0, 0)$. The Jacobian here reads

$$\mathbf{J}(1, 0, 0) = \begin{bmatrix} -1/s & -1/s & 0 \\ 0 & 0 & 1 \\ 0 & a(b-1) & s \end{bmatrix} \quad (4.48)$$

giving the three eigenvalues

$$\lambda_1 = -\frac{1}{s} \quad \lambda_{2,3} = \frac{1}{2} \left(s \pm \sqrt{s^2 + 4a(b-1)} \right) \quad (4.49)$$

We seek real eigenvalues. Since $b - 1 \leq 0$ we have that

$$s^2 + 4a(b-1) \geq 0 \text{ implying } s^2 - 4|a(b-1)| \geq 0, \quad (4.50)$$

and therefore, the minimum allowed speed is given by $s_{\min} = 2\sqrt{|a(b-1)|}$. $\diamond$

Question How can we interpret the reaction terms in Equation (4.43) in context of a predator-prey system? What is the biological interpretation of the parameters a and b ?

Exercise 34 Perform a perturbation analysis of the Fisher-Kolmogorov front up to first order in ϵ ; you can express c_1 in terms of c_0 . Hints:

1. Show that $1 - 2c_0 = -c''_0/c'_0$.
2. Then show that equation for the first order term is

$$\frac{dc_1}{d\xi} - \frac{c''_0}{c'_0} c_1 = -kD \frac{d^2 c_0}{d\xi^2}.$$

3. Solve this using the general identity from calculus

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)|.$$

4. Finally, use the initial value $u'_0(0) = 1/4$ to find the particular solution.

Exercise 35 Compare the zeroth order perturbation result, Equation (4.42), with numerical results for different front speeds.

Exercise 36 Recall, our imposed conditions for a front to exist. Can the linear problem

$$\frac{\partial c}{\partial t} = c + \frac{\partial^2 c}{\partial x^2} \quad (4.51)$$

feature a front?

Exercise 37 Through linear analysis of Equation (4.47), show that the two-species predator-prey model can feature (fascinating) spatial oscillations in concentrations in the front zone for certain choices of parameters.

Exercise 38 In Example 20 the predator featured mobility. Will the model support a propagating front if the predator is immobile, but the prey is mobile?

Exercise 39 The spread of dominant genes is modelled through an extended Fisher-Kolmogorov wavefront

$$\frac{\partial N}{\partial t} = k(N_0 - N)N - N + D \frac{\partial^2 N}{\partial x^2}. \quad (4.52)$$

$N = N(x, t) \geq 0$ represents the dominant gene concentration, k , N_0 , and D are the rate constant, total gene pool, and diffusion coefficient, respectively. We let $x > 0$ and $t \geq 0$.

We assume that Equation (4.52) allows for a propagating wave with constant speed s .

1. Perform a coordinate transformation using $z = x - st$ such that the wave is fixed in this coordinate system.

2. Show that $N \geq 0$ implies $kN_0 > 1$ if the wave front exists. What is the constant gene concentration behind the front?
3. Show that the minimum allowed speed is $s_{\min} = 2\sqrt{D(kN_0 - 1)}$ if the front exists.
4. Perform numerical simulations confirming the results above. Describe what happens when $k = 1$, $D = 1$ and $N_0 = 0.5$ (ignoring coefficient units). Clearly state your choice of boundary conditions.

Exploration 5 One main result from our study of the Fisher-Kolmogorov front is that $s \geq s_{\min} = 2\sqrt{kD}$. As stated in the text, we do not know *if* such a front exists, and if so *what* s actually is. (We only know the minimum speed allowed.) Perform a numerical exploration of the system, and confirm whether the front exists (under different ICs). If so determine the stable front speed. [Note: Carefully explore the convergence of the wave speed.]

4.3 Turing structures

Recall Example 3 from the Introduction; in this section we will return to *diffusion-driven spatial structures* and investigate how they emerge in reaction-diffusion systems. As a first thought, the existence of apparently spontaneously formed structures contradicts our intuition about diffusion processes which tend to reduce and eventually remove gradients. The structures were first found theoretically by *Alan Turing*¹ in the early 1950s, which is why they also bear his name.

We will here explore the Turing structures through the *FitzHugh-Nagumo (FHN) model* used to study excitable systems. For example, it is a standard model for neuron electrical signals caused by voltage build-up and release (also called neuron firing). A simple version of the model is in the homogeneous case where we can ignore diffusion

$$\frac{dc_1}{dt} = c_1 - c_1^3 - c_2 + I = r_1(c_1, c_2) \quad (4.53)$$

$$\frac{dc_2}{dt} = \epsilon(c_1 - c_2) = r_2(c_1, c_2) \quad (4.54)$$

c_1 is the voltage and c_2 acts as a variable that describes the coupling between the voltage and other relevant cell mechanisms. Notice the definitions of r_1 and r_2 on the right-hand sides; they will be used below. The model parameters satisfy $I, \epsilon > 0$. For now, we will not go into a more detailed bio-physical interpretation of the model, and right away start the mathematical analysis.

It is easy to verify that there exists a single fixed point, (a, b) , namely,

$$(a, b) = (I^{1/3}, I^{1/3}) . \quad (4.55)$$

¹Who, by the way, is also famous for inventing the Turing machine and cracking the Enigma code during the Second World War.

We will use a and b for the fixed point coordinates to ease and clarify the reading a bit. It is worth to go through some of the details we know from standard linear analysis: We first linearise the dynamical equations in a neighbourhood around the fixed point (a, b) . For r_1 we have

$$r_1 = r_1(a, b) + \frac{\partial r_1}{\partial c_1}(c_1 - a) + \frac{\partial r_1}{\partial c_2}(c_2 - b) + \dots \quad (4.56)$$

$$= (1 - 3a^2)u - v + \dots \quad (4.57)$$

where u and v describe the deviation of c_1 and c_2 from the fixed point, that is,

$$u = c_1 - a \quad \text{and} \quad v = c_2 - b. \quad (4.58)$$

For r_2 we, of course, have the linear function

$$r_2 = \epsilon(u - v). \quad (4.59)$$

We now write the Jacobian in the fixed point in terms of the original parameters I and ϵ

$$\mathbf{J} = \begin{bmatrix} \gamma & -1 \\ \epsilon & -\epsilon \end{bmatrix}, \quad (4.60)$$

where $\gamma = 1 - 3I^{2/3}$ just acts a simple placeholder. The eigenvalues are found from the characteristic polynomial

$$\lambda^2 + (\epsilon - \gamma)\lambda + \epsilon(1 - \gamma) = 0. \quad (4.61)$$

giving

$$\lambda_{1,2} = \frac{1}{2} \left(\gamma - \epsilon \pm \sqrt{(\epsilon - \gamma)^2 - 4\epsilon(1 - \gamma)} \right). \quad (4.62)$$

We now focus on the case where we have complex conjugate eigenvalues, that is, $4\epsilon > (\epsilon + \gamma)^2$. We then have that the fixed point is

$$\begin{array}{ll} \text{stable if} & \gamma < \epsilon \\ \text{non-hyperbolic if} & \gamma = \epsilon \\ \text{unstable if} & \gamma > \epsilon \end{array}$$

In fact, the system undergoes a Hopf bifurcation at $\gamma = \epsilon$. To show this we keep I fixed, i.e., fixing γ , and let ϵ act as bifurcation parameter. The functions r_1 and r_2 are C^∞ , $\text{Re}(\lambda_{1,2}) = 0$ for $\epsilon = \gamma$, and

$$\frac{d\text{Re}(\lambda_{1,2})}{d\epsilon} = -1/2 \neq 0 \quad (4\epsilon > (\epsilon + \gamma)^2) \quad (4.63)$$

This last property is *the transversal condition* which ensures that the eigenvalue real part, in fact, changes sign.

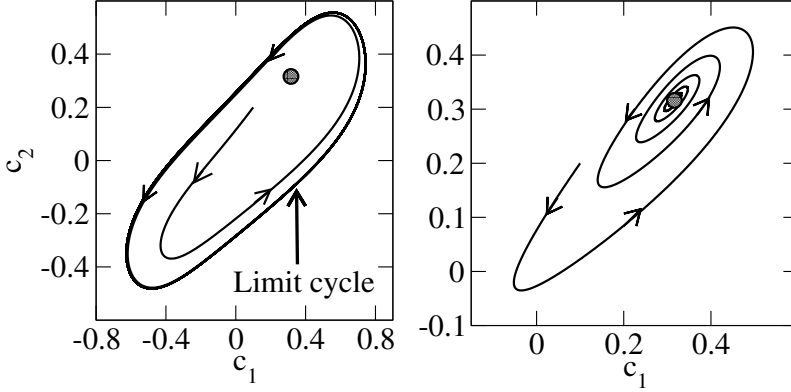


Figure 4.5: Phase plane trajectories for the FHN model system. $I^{2/3} = 1/10$. (a) $\epsilon < 7/10$. (b) $\epsilon > 7/10$. Fixed point illustrated by the filled circle. Trajectories are obtained from numerical solutions.

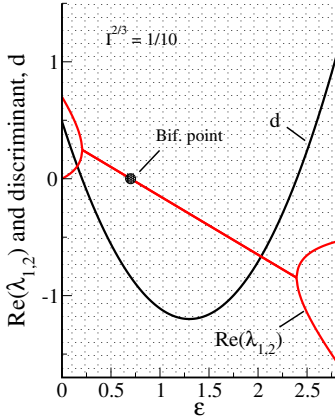


Figure 4.5 shows the phase plane of the FHN model before and after the bifurcation.

We now include diffusion for both c_1 and c_2 and show that if the diffusion coefficients are different, $D_1 \neq D_2$, the system may feature an instability even if it is stable in the homogeneous case. Without compromising the main point, we can choose values

$$I^{2/3} = 1/10 \quad \text{and} \quad \epsilon = 1, \quad (4.64)$$

which corresponds to a stable system in the absence of diffusion.

First, notice that

$$\frac{\partial c_1}{\partial t} = \frac{\partial u}{\partial t} \quad \text{and} \quad \frac{\partial^2 c_1}{\partial x^2} = \frac{\partial^2 u}{\partial x^2} \quad (4.65)$$

and so forth for the derivatives of c_2 and v . Then we can write the linearised RD-equations in terms of u and v , i.e., in terms of the deviation from the homogeneous fixed point

$$\frac{\partial u}{\partial t} = \gamma u - v + D_1 \frac{\partial^2 u}{\partial x^2} \quad (4.66)$$

$$\frac{\partial v}{\partial t} = u - v + D_2 \frac{\partial^2 v}{\partial x^2} \quad (4.67)$$

with D_1 and D_2 being the usual diffusion coefficients. The boundaries we apply here are the zero Neumann BCs

$$\left. \frac{\partial u}{\partial x} \right|_{x=0} = \left. \frac{\partial u}{\partial x} \right|_{x=L} = \left. \frac{\partial v}{\partial x} \right|_{x=0} = \left. \frac{\partial v}{\partial x} \right|_{x=L} = 0. \quad (4.68)$$

The initial condition is such that the spatial averages of u and v are zero

$$\frac{1}{L} \int_0^L u(x, 0) dx = 0 \text{ and } \frac{1}{L} \int_0^L v(x, 0) dx = 0, \quad (4.69)$$

but we allow u and v some random distribution around the fixed point with sufficiently small amplitude.

From our earlier experience with the linear RD-equation under Neumann BCs we expect that the solutions for u and v are Fourier cosine series, that is, we propose that

$$u(x, t) = \sum_{n=1}^{\infty} u_n e^{\lambda_n t} \cos\left(\frac{n\pi}{L} x\right) \text{ and } v(x, t) = \sum_{n=1}^{\infty} v_n e^{\lambda_n t} \cos\left(\frac{n\pi}{L} x\right), \quad (4.70)$$

where u_n and v_n are Fourier coefficients in the series.

Question In general, the cosine Fourier series includes a constant term. Why is there no constant term in Equation (4.70)?

We do not know λ_n , that is, in our proposition above the temporal dynamics is still undetermined. Now, the solutions must fulfill the two linearised equations. Substituting the guess, Equation (4.70), into Equation (4.66) and differentiating we get for each term, n ,

$$\begin{aligned} \lambda_n u_n e^{\lambda_n t} \cos\left(\frac{n\pi}{L} x\right) &= \frac{7}{10} u_n e^{\lambda_n t} \cos\left(\frac{n\pi}{L} x\right) \\ &\quad - v_n e^{\lambda_n t} \cos\left(\frac{n\pi}{L} x\right) - D_1 u_n \left(\frac{n\pi}{L}\right)^2 e^{\lambda_n t} \cos\left(\frac{n\pi}{L} x\right) \end{aligned} \quad (4.71)$$

The exponential terms cancels, and rearranging we obtain

$$\left(\lambda_n u_n - \frac{7}{10} u_n + v_n + \left(\frac{n\pi}{L}\right)^2 u_n D_1 \right) \cos\left(\frac{n\pi}{L} x\right) = 0. \quad (4.72)$$

Repeating this for Equation (4.67) we get the identity

$$\left(\lambda_n v_n - u_n + v_n + \left(\frac{n\pi}{L}\right)^2 v_n D_2 \right) \cos\left(\frac{n\pi}{L} x\right) = 0. \quad (4.73)$$

We can combine these two equations and obtain a more convenient (and hopefully recognizable) form. First, let $\mathbf{W}_n = (u_n, v_n)^T$ and recall that the diffusion matrix is defined as

$$\mathbf{D} = \begin{bmatrix} D_1 & 0 \\ 0 & D_2 \end{bmatrix} \quad (4.74)$$

then we have in matrix-vector form

$$\left[\lambda_n \mathbf{I} - \mathbf{J} + \left(\frac{n\pi}{L}\right)^2 \mathbf{D} \right] \cdot \mathbf{W}_n = \mathbf{0}. \quad (4.75)$$

This, you may see, is an eigenvalue problem. As usual, we do not seek the trivial case $\mathbf{W}_n = \mathbf{0}$, as this leads to the trivial solutions for u_n and v_n . A non-trivial solution exists if the matrix

$$\left[\lambda_n \mathbf{I} - \mathbf{J} + \left(\frac{n\pi}{L} \right)^2 \mathbf{D} \right] \quad (4.76)$$

is rank deficient, that is, if there exists one or two λ_n such that the determinant is zero

$$\left| \lambda_n \mathbf{I} - \mathbf{J} + \left(\frac{n\pi}{L} \right)^2 \mathbf{D} \right| = \left(\lambda_n - \frac{7}{10} + \left(\frac{n\pi}{L} \right)^2 D_1 \right) \left(\lambda_n + 1 + \left(\frac{n\pi}{L} \right)^2 D_2 \right) + 1 = 0. \quad (4.77)$$

Again, solving this algebraic problem is a rather tedious task. To ease the reading, we therefore introduce the two auxiliary variables

$$a_n = -\frac{7}{10} + \left(\frac{n\pi}{L} \right)^2 D_1, \text{ and } b_n = 1 + \left(\frac{n\pi}{L} \right)^2 D_2. \quad (4.78)$$

The solution can now formally be written as

$$\lambda_{n,1,2} = \frac{1}{2} \left(-(a_n + b_n) \pm \sqrt{(a_n + b_n)^2 - 4(1 + a_n b_n)} \right). \quad (4.79)$$

Notice that the eigenvalues are functions of n , D_1 , and D_2 .

We can now investigate the dynamics for each Fourier term (or Fourier mode): If the real part of the eigenvalue is negative for a given mode this is stable, whereas if the real part is positive the mode is unstable. As always, in case of zero real part (the non-hyperbolic case) we cannot conclude anything about the stability of the system. Figure 4.6 shows the real part of the eigenvalues λ_n as function of wave vector, which is equivalent to a plot of the real part of λ_n versus n . This is the *dispersion plot*, and the eigenvalue's dependency on k is called a *dispersion relation*. We have used $D_2 = 10D_1$. Some of the Fourier modes (indicated by the blue interval) are unstable, hence, the solution to the linearized system, Equation (4.70), grows exponentially in time and the system is unstable. If just one mode features this instability the system is unstable.

This is a linear analysis. The non-linear part of the system may eventually suppress the exponential growth as u and v become sufficiently large and thus form a steady state structure. We will not explore such nonlinear features mathematically, however, we can solve the non-linear RD-equation using the FTCS algorithm. We will do this using the same parameter values as in Fig. 4.6. The voltage c_1 is plotted as function of x for four different times in Fig. 4.7. It is seen that the structure emerges from the homogeneous state reaching a final steady state. An important observation from the numerical simulations is that the (characteristic) period of the structure is given by

$$2\pi/k_{\max}. \quad (4.80)$$

$k_{\max}$ is also shown in Fig. 4.6. This means that the Fourier mode with largest eigenvalue (real part) grows the fastest and dominates the final steady state. Again, this is all based on the linear analysis and not something we can conclude in general.

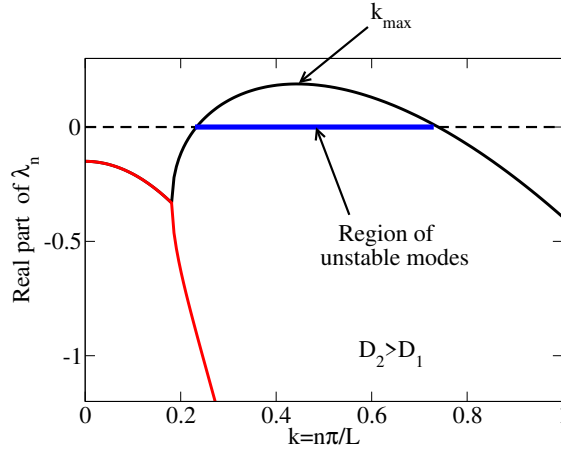


Figure 4.6: Dispersion plot: Real part of the eigenvalues, λ_n as a function of $k = n\pi/L$. $D_2 = 10D_1$, $\alpha = (1/10)^{3/2}$, and $\epsilon = 1$.

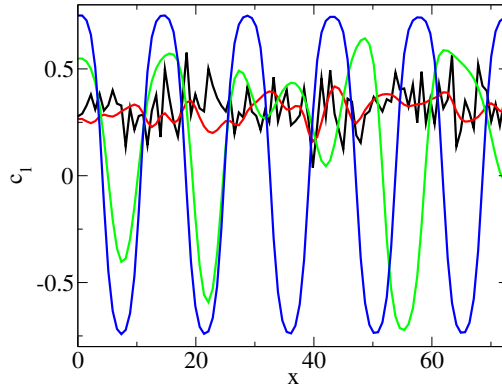


Figure 4.7: FHN model voltage profiles as function of x for four different times. Black line is the condition. Red and green lines are transient profiles, and blue is the final steady state profile. Parameter values are the same as in Fig. 4.6.

We list three necessary conditions for a system to feature Turing structures.

Theorem Consider a two species RD-equation with equilibrium point (a, b) and Jacobian

$$\mathbf{J}(a, b) = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad (4.81)$$

If the system features Turing structures then

1. $a_{11}a_{22} < 0$
2. $a_{12}a_{21} < 0$
3. $D_1 \neq D_2$.

◦

Please notice that these are necessary conditions, but do not guarantee a diffusion-driven instability. We will not prove this theorem.

Example 21 Let $a \in \mathbb{R}_+$ be a parameter then the Jacobian matrix

$$\mathbf{J} = \begin{bmatrix} -2a & -a \\ 4 & a \end{bmatrix} \quad (4.82)$$

fulfills the necessary criterion for Turing patterns, but

$$\mathbf{J} = \begin{bmatrix} -2a & a \\ 4 & a \end{bmatrix} \quad (4.83)$$

does not. ◇

We end this section by summarising the key concepts in

1. We consider a nonlinear system of at least two dynamical variables.
2. The system must possess a stable fixed point when diffusion is absent.
3. Diffusion processes are introduced, $D_1 \neq D_2$, which can lead to excitation of spatial Fourier modes.
4. The nonlinear system dynamics may subsequently produce a stationary spatial pattern.

Exercise 40 For the FitzHugh-Nagumo model use $D_1 = 1, \alpha = (1/10)^{3/2}, \epsilon = 2$, and $L = 200$. Find the values for D_2 and corresponding wave modes n , where unstable Fourier modes emerge. Predict the wavelength, and confirm your findings from numerical simulations of the RD-eq system.

Exercise 41 Theorem 4.3 limits the possibilities of signs the Jacobian matrix elements can have. Illustrate this by writing the Jacobian with the elements sign only. (This leads to two concepts, namely, *activator-inhibitor* and *positive feedback*.)

Exercise 42 Recall the Lotka-Volterra predator-prey model

$$\begin{aligned}\frac{dc_1}{dt} &= c_1 - c_1 c_2 \\ \frac{dc_2}{dt} &= c_1 c_2 - \alpha c_2\end{aligned}\tag{4.84}$$

with $\alpha > 0$ and $c_1, c_2 \geq 0$. Does the corresponding RD-equation allow for diffusion-driven instabilities?

Exercise 43 We consider the RD-equation system

$$\frac{\partial c_1}{\partial t} = \frac{c_1^2}{c_2} - bc_1 + \frac{\partial^2 c_1}{\partial x^2}\tag{4.85}$$

$$\frac{\partial c_2}{\partial t} = c_1^2 - c_2 + D \frac{\partial^2 c_2}{\partial x^2}\tag{4.86}$$

Determine the fixed point for $c_1, c_2 > 0$ in the homogeneous case. Then determine the critical value of b , where this fixed point is stable. Now, choose $b = 0.5$, and find the dispersion curve for different values of D ; identify the range of possible unstable Fourier modes on the curve. Based on this analysis, explore numerically whether Turing structures exist. (Inspired by Murray, *Mathematical Biology*.)

Chapter 5

Appendices

5.1 The slow Fourier transform (SFT)

Recall that the Fourier coefficients for a function f are given by

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos(n\pi x/L) dx \quad (5.1)$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin(n\pi x/L) dx \quad (5.2)$$

These two integrals are on the general form $\int_{-L}^L g(x)dx$. Now, integrals can be quite a mouthful, and it is therefore useful to devise a numerical and therefore approximative method.

The geometrical interpretation of an integral for a function $g : \mathbb{R} \supseteq D \rightarrow \mathbb{R}$ is the area between the first axis and the graf of g (with sign!). The area can be approximated by rectangles; this is also the fundamental idea behind both Riemann and Darboux definitions of the integral. Figure 5.1 illustrates how a set of rectangular areas $A_1, A_2, \dots$ approximates the integral of g in the interval $x_0 \leq x \leq x_N$. Each area is

$$A_i = (x_i - x_{i-1}) \frac{g(x_i) + g(x_{i-1})}{2} \quad (5.3)$$

and we hope that

$$\int_{x_0}^{x_N} g(x)dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n A_i. \quad (5.4)$$

This is the trapezoidal rule.

Exercise 44 Implement the trapezoidal rule. Test it against a known integral, for example, $\int_0^{2\pi} \cos(x)dx$. Explore the trapz rule's precision by changing the sub-interval length $x_i - x_{i-1}$.

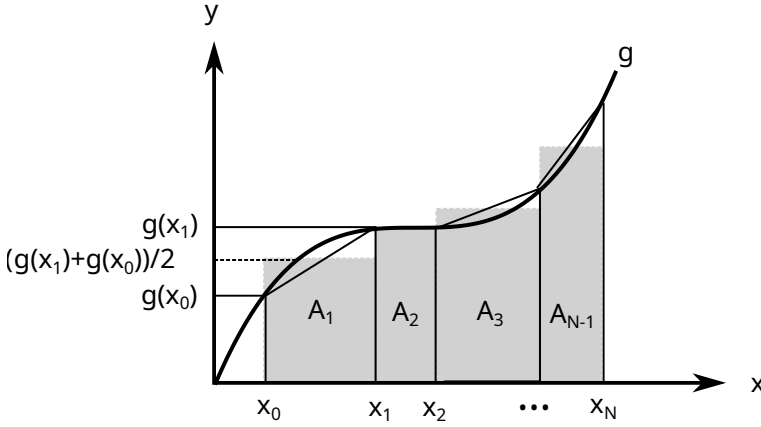


Figure 5.1: Illustration of the trapezoidal rule

Exercise 45 Let $f(x) = -x$, where for $-1 \leq x \leq 1$ such that $L = 1$. Use your implementation of the trapezoidal rule to numerically calculate the Fourier coefficients and compare with the analytical result.

Exercise 46 Download the data file `fourier.dat` from the moodle page. How many different Fourier modes can you detect using your code?

5.2 The FTCS algorithm

Before advising a numerical scheme for the diffusion equation it is illustrative to first see an example a scalar differential equation, and how this can be solved numerically. In particular, it will introduce some of the symbolism we will use later.

We consider the general first order problem

$$\frac{dc}{dt} = c'(t) = f(c, t), \quad c(t_0) = c_0 \quad (5.5)$$

such that $c : I \rightarrow \mathbb{R}$, $I \subseteq \mathbb{R}$ and f is known, of course. This defines an *initial value problem*; for these problems we usually use t as the free variable rather than x .

We assume that the function c can be represented by a Taylor series, i.e., by definition c is analytical in all interior points in I . (We actually do not need this strict condition, but it will save us some headaches.) We then Taylor expand c around an interior point $a \in I$

$$c(t) = c(a) + c'(a)(t - a) + \sum_{k=2}^{\infty} \frac{c^{(k)}(a)}{k!} (t - a)^k. \quad (5.6)$$

We will write this in a slightly different way by introducing $\Delta t = t - a$

$$\begin{aligned} c(a + \Delta t) &= c(a) + c'(a)\Delta t + \sum_{k=2}^{\infty} \frac{c^{(k)}(a)}{k!} \Delta t^k \\ &= c(a) + c'(a)\Delta t + \mathcal{O}(\Delta t^2) \end{aligned} \quad (5.7)$$

where $\Delta t = t - a$. $\mathcal{O}$ is the remainder function (called big-O) and the argument to $\mathcal{O}$ indicates the lowest order in Δt , here Δt^2 . Loosely speaking, we say that $\mathcal{O}(\Delta t^k)$ goes faster to zero than $\mathcal{O}(\Delta t^{k-1})$ as Δt goes to zero. Re-arranging gives

$$c'(a) = \frac{c(a + \Delta t) - c(a)}{\Delta t} - \mathcal{O}(\Delta t) \quad (5.8)$$

using $\mathcal{O}(\Delta t^2)/\Delta t = \mathcal{O}(\Delta t)$. Substitution into the original differential equation, Eq. (5.5), and re-arranging gives

$$c(a + \Delta t) = c(a) + \Delta t f(c(a), a) + \mathcal{O}(\Delta t^2). \quad (5.9)$$

To use this in practise we can let $t_0 = 0$, that is, we have the initial condition $c(0) = c_0$. Then, we have by substitution

$$\begin{aligned} c(\Delta t) &= c_0 + \Delta t f(c_0, 0) + \mathcal{O}(\Delta t^2) \\ c(2\Delta t) &= c(\Delta t) + \Delta t f(c(\Delta t), \Delta t) + \mathcal{O}(\Delta t^2) \\ &\dots \\ c((n+1)\Delta t) &= c(n\Delta t) + \Delta t f(c(n\Delta t), n\Delta t) + \mathcal{O}(\Delta t^2) \\ &\dots \end{aligned} \quad (5.10)$$

This presents an iterative scheme: from $c(0) = c_0$ we evaluate the function value at $t = \Delta t$, and from this the value at $t = 2\Delta t$, etc. This is the simplest numerical scheme we can compose, and it is called the *first order forward Euler scheme* or just the Euler scheme.

Usually we replace the argument with the iteration index such that $c(n\Delta t) = c^n$. Eq. (5.10) is then written as (leaving out the $\mathcal{O}$ -function)

$$c^{n+1} = c^n + \Delta t f(c^n, n\Delta t) \quad (5.11)$$

The Euler scheme is called an explicit scheme because the value at time $(n+1)\Delta t$ is given explicitly in terms of values at time $n\Delta t$. It is important to note that explicit schemes are so called non-symplectic and they cannot be used to solve problems where there is a system constant, for example, a constant energy.

Moving on to the diffusion equation. Recall, for a Dirichelt problem we have

$$\frac{\partial c}{\partial t} = \frac{\partial^2 c}{\partial x^2} \quad (5.12)$$

with boundary and initial values

$$c(x, 0) = f(x) \quad \text{and} \quad c(0, t) = c_0 \quad c(L, t) = c_L. \quad (5.13)$$

Again c is analytical on its domain, $t \geq 0$, and $0 \leq x \leq L$.

Keeping x fixed we can Taylor expand with respect to t giving

$$c(x, a + \Delta t) = c(x, a) + \left. \frac{\partial c}{\partial t} \right|_{t=a} \Delta t + \mathcal{O}(\Delta t^2), \quad (5.14)$$

or equivalently

$$\left. \frac{\partial c}{\partial t} \right|_{t=a} = \frac{c(x, a + \Delta t) - c(x, a)}{\Delta t} - \mathcal{O}(\Delta t). \quad (5.15)$$

Now, keep t fixed. We Taylor expand up to second order with respect to x , thus, around a point $0 < b < L$ we get

$$c(b + \Delta x, t) = c(b, t) + \left. \frac{\partial c}{\partial x} \right|_{x=b} \Delta x + \frac{1}{2} \left. \frac{\partial^2 c}{\partial x^2} \right|_{x=b} \Delta x^2 + \mathcal{O}(\Delta x^3) \quad (5.16)$$

where $\Delta x = x - b$. There is an annoying first order derivative here; thankfully we can eliminate this by noting that

$$c(b - \Delta x, t) = c(b, t) - \left. \frac{\partial c}{\partial x} \right|_{x=b} \Delta x + \frac{1}{2} \left. \frac{\partial^2 c}{\partial x^2} \right|_{x=b} \Delta x^2 + \mathcal{O}(\Delta x^3) \quad (5.17)$$

Hence, addition of Eqs.(5.16) and (5.17) gives

$$c(b + \Delta x, t) + c(b - \Delta x, t) = 2c(b, t) + \left. \frac{\partial^2 c}{\partial x^2} \right|_{x=b} \Delta x^2 + \mathcal{O}(\Delta x^4) \quad (5.18)$$

or

$$\left. \frac{\partial^2 c}{\partial x^2} \right|_{x=b} = \frac{c(b + \Delta x, t) - 2c(b, t) + c(b - \Delta x, t)}{\Delta x^2} - \mathcal{O}(\Delta x^2) \quad (5.19)$$

Again, we replace the arguments with iteration index, i.e.,

$$\begin{aligned} c(x, t) &= c_i^n \\ c(x, t + \Delta t) &= c_i^{n+1} \\ c(b + \Delta x, t) &= c_{i+1}^n \\ c(b - \Delta x, t) &= c_{i-1}^n \end{aligned} \quad (5.20)$$

and we write the diffusion equation as (leaving out the $\mathcal{O}$ s)

$$\frac{c_i^{n+1} - c_i^n}{\Delta t} \approx \frac{c_{i-1}^n - 2c_i^n + c_{i+1}^n}{\Delta x^2} \quad (5.21)$$

readily giving the iterative scheme for point i

$$c_i^{n+1} = c_i^n + \Delta t \left[\frac{c_{i-1}^n - 2c_i^n + c_{i+1}^n}{\Delta x^2} \right]. \quad (5.22)$$

This scheme is called the forward time centered space (FTCS) algorithm and is also an explicit scheme.

Exercise 47 Consider the differential equation

$$c'(t) = 1 - c \text{ with } c(0) = 0 \quad (5.23)$$

Solve this initial value problem. Then implement the Euler scheme and solve the problem numerically. (Use your favorite programming language; if this is Java you are on your own!) Test your implementation against the analytical result.

Note: The error we make at each time step can be evaluated directly from the higher order Taylor polynomial terms if the function c is known. Use this fact if you wish.

Exercise 48 Let $c(x) = 1 - x^2$, where $0 \leq x \leq 1$ and $c(0) = c(1) = 0$. Use Eq. (5.19) to calculate the second order derivative. Compare to the analytical result. Make a plot of the error as function of Δx .

Exercise 49 Implement the FTCS algorithm. Use this to solve a diffusion problem where the analytical solution is known. Discuss possible errors.

5.3 Existence and uniqueness of the RD-equation

We here list the general criteria for a solution to the RD-equation to exist; again we will not prove these criteria, but discuss examples.

First, we must define Lipschitz continuity: A function $f : I \rightarrow \mathbb{R}$ is Lipschitz continuous on the real interval I if there exists a $K \in \mathbb{R}$ such that

$$|f(x) - f(y)| \leq K|x - y|, \quad x, y \in I. \quad (5.24)$$

This definition basically states that if the function is Lipschitz continuous, the function cannot have an infinite rate of change anywhere on its domain. Let see a standard example of function, which is *not* Lipschitz continuous.

Example 22 The function $f(x) = \sqrt{|x|}$ is not Lipschitz continuous on the domain $I = [-1, 1]$.

Proof We focus on the likely troublesome point $x = 0$; we have from the definition

$$|f(0) - f(y)| \leq K|0 - y| \Rightarrow |f(y)| \leq K|y|, \quad (5.25)$$

that is, for $y \neq 0$ we get

$$K = \frac{\sqrt{|y|}}{|y|} = \frac{1}{\sqrt{|y|}}. \quad (y \neq 0) \quad (5.26)$$

This fraction diverges as $y \rightarrow 0$, hence, K does not exist. •

Notice that f is continuous in the usual Weierstrass definition and that Lipschitz continuity is in this sense a "stronger" condition on the function than usual continuity. $\diamond$

We are now ready to list the four criteria that guarantee the existence and uniqueness of a solution to the RD-equation

$$\frac{\partial c}{\partial t} = r(c) + D \frac{\partial^2 c}{\partial x^2} , \quad (5.27)$$

with IC $c(x, 0) = f(x)$ and Dirichlet BCs

$$c(0, t) = c(0, L) = 0 , \quad (5.28)$$

or Neumann BCs

$$\left. \frac{\partial c}{\partial x} \right|_{x=0} = \left. \frac{\partial c}{\partial x} \right|_{x=L} = 0 , \quad (5.29)$$

where $x \in [0; L]$ and $t \geq 0$. These are (from Allen *An Introduction to Mathematical Biology*),

1. $f(x)$ is continuous on the interior open domain $]0; L[$,
2. $f(x)$ has a lower and upper bound, i.e., $\exists a, b \in \mathbb{R} : a \leq f(x) \leq b$ where $x \in]0; L[$,
3. the reaction function fulfills $r(a) \geq 0$ and $r(b) \leq 0$, and
4. r is Lipschitz continuous on $[0; L]$

Let us at least see an example.

Example 23 Let us examine the existence and uniqueness for Example 19 on page 58 using $k > 0$. (i) $f(x)$ is continuous for all $x \in]0; L[$. (ii) Since $0 \leq f(x) \leq 1$, f has a lower and an upper bound. (iii) $r(0) = 0$, but since $r(1) \leq 0$ we are not guaranteed a solution. This is consistent with the discussion in the example. $\diamond$

Question Are we guaranteed a solution for Example 19 if $k < 0$?

Exercise 50 Are we guaranteed a solution to the Fisher-Kolmogorov equation

$$\frac{\partial c}{\partial t} = kc(1 - c) + D \frac{\partial^2 c}{\partial x^2} \quad (5.30)$$

with zero Neumann BCs if the IC fulfills $0 \leq f(x) \leq 1$ and is continuous on $]0; 1[$?